133

Genital and Lower Urinary Tract Trauma

Allen F. Morey, and Jay Simhan

Key words

bladder; penis; scrotum; trauma; urethra

Questions

1. Which of the following is an absolute indication for open repair of blunt bladder rupture injury?
    a. Significant extraperitoneal bladder rupture with extravasation of contrast agent into the scrotum
    b. Significant extraperitoneal bladder rupture with gross hematuria
    c. Significant extraperitoneal bladder rupture that has not healed after 3 weeks of Foley catheter drainage
    d. Intraperitoneal bladder rupture
    e. Significant extraperitoneal bladder rupture associated with pelvic fracture requiring treatment by external fixation
2. Which of the following statements is TRUE regarding cystography for diagnosis of bladder injury?
    a. If the patient is already undergoing computed tomography (CT) for evaluation of associated injuries, CT cystography should be performed via antegrade filling of the bladder after

intravenous administration of radiographic contrast material and clamping the Foley catheter.

b. If plain film cystograms are obtained, the study is considered negative and complete if there is no extravasation of contrast agent seen on the filling film.

c. CT cystography is best performed with undiluted contrast medium.

d. An absolute indication for immediate cystography is the presence of pelvic fracture and microhematuria.

e. None of the above.

3. Which of the following statements is TRUE about blunt bladder rupture injuries?

a. They are present in 90% of patients presenting with pelvic fractures.

b. They coexist with urethral disruption in 50% of cases.

c. Extraperitoneal ruptures are always amenable to nonoperative treatment.

d. High mortality rate is primarily related to nonurologic comorbidities.

e. They are associated with microhematuria or no hematuria in 40% of cases.

4. The risk of complications from nonoperative treatment of extraperitoneal bladder rupture is increased by:

a. associated orthopedic injury.

b. associated vaginal injury.

c. associated urethral injury.

d. associated rectal injury.

e. all of the above.

5. Three months after a urethral distraction injury, a patient is found to have a 2-cm obliterative posterior urethral defect. Which of the following is TRUE about the repair?

a. One-stage, open, perineal anastomotic urethroplasty is preferred.

b. Orthopedic hardware in the pubic symphysis area is a contraindication to open posterior urethroplasty.

c. Buccal mucosa graft urethroplasty is recommended.

d. Urethral stent placement is recommended.
e. The patient is at high risk for incontinence after posterior urethral reconstruction surgery.

6. In a patient with a pelvic fracture from blunt trauma in whom no urine is returned after catheter placement, what is the best initial method to evaluate urethral injury?
   a. Retrograde urethrography
   b. CT of abdomen and pelvis
   c. Filiforms and followers
   d. Bladder ultrasonography
   e. None of the above
7. During exploration after a scrotal gunshot wound, 20% of the left testicular capsule is found to be disrupted. What should be done?
   a. Left orchiectomy
   b. Application of wet dressings and delayed testicular surgery
   c. Left testicular reconstruction with synthetic graft
   d. Closure of the scrotal laceration followed by ultrasonography
   e. Immediate primary repair of the left testis
8. A 23-year-old man is found to have an 80% transection of the proximal bulbar urethra after a gunshot wound with a 22-caliber pistol. A 1-cm urethral defect is visualized during cystoscopy. What is the most appropriate therapy?
   a. Buccal mucosa graft urethroplasty
   b. Spatulated, stented, tension-free, watertight repair of the urethra with absorbable sutures
   c. Suprapubic tube placement
   d. Urethral catheterization alone
   e. Perineal urethrostomy
9. Which of the following statements regarding penile fracture is FALSE?
   a. Most injuries occur ventrolaterally.
   b. Rupture of a superficial vein can sometimes mimic the presentation of a corporal tear.
   c. Retrograde urethrography should be uniformly performed to assess for urethral injury.

d. Patients with penile fracture who are treated nonoperatively are more likely to have longer hospital stays, a higher risk of infection, and penile curvature than those whose fracture is repaired surgically.
e. Physical examination is usually sufficient in making the diagnosis or for deciding on surgical exploration.

10. The blood in a hematocele is contained in which of the following?
    a. Tunica albuginea
    b. Tunica vaginalis
    c. Dartos muscle
    d. Camper fascia
    e. Spermatic cord
11. Blunt scrotal trauma that results in testis rupture:
    a. is usually a bilateral process.
    b. is often diagnosed by the presence of intratesticular hypoechoic areas on ultrasonography.
    c. has a degree of hematoma that correlates with the extent of injury.
    d. requires conservative management that results in acceptable viability and function.
    e. is definitively diagnosed during physical examination alone in most cases.
12. Which of the following statements is TRUE regarding penile amputation injury?
    a. Microscopic reanastomosis of the corporal arteries is recommended.
    b. The severed phallus should be placed directly on ice during transport.
    c. Microscopic dorsal vascular and neural reanastomosis is the best method of repair.
    d. Primary macroscopic reanastomosis invariably results in erectile dysfunction.
    e. Skin loss is rarely a problem after macroscopic repair.
13. What is the best option for coverage of acute penile skin loss?
    a. Foreskin flap for small distal lesions
    b. Meshed skin graft in a young child

c. Wet-to-dry dressings
d. Thigh flaps
e. Burying the penile shaft in a scrotal skin tunnel

14. Which is FALSE about penile fracture?
    a. Penile fracture must be repaired immediately for the best outcomes.
    b. Ultrasonography can identify location of the corporal tear.
    c. Magnetic resonance imaging (MRI) can demonstrate disruption of the tunica albuginea.
    d. Rupture of the dorsal penile artery can have the same presentation as penile fracture.
    e. Bilateral corporal injury is more commonly associated with urethral injury.
15. Advantages of open suprapubic tube placement after posterior urethral disruption injuries include:
    a. inspection of bladder.
    b. an opportunity for controlled antegrade urethral realignment.
    c. allowance for large-bore catheter insertion.
    d. not jeopardizing continence or potency rates.
    e. all of the above.

Answers

1. **d. Intraperitoneal bladder rupture.** When intraperitoneal bladder laceration occurs after blunt trauma, a large laceration of the bladder dome is usually produced that predisposes to urinary ascites and/or peritonitis if it is not repaired promptly.
2. **e. None of the above.** The CT cystogram must be performed via retrograde distention of the bladder with a diluted contrast medium. Most bladder lacerations are associated with gross hematuria, not microhematuria. A drainage film is required to complete a plain film cystogram.
3. **d. High mortality rate is primarily related to nonurologic comorbidities.** Bladder lacerations occur in approximately 10% of

pelvic fractures and often occur in the context of multisystemic trauma.

4. **e. All of the above.** All of the listed concomitant injuries increase the risk of complications such as abscess, fistula, or incontinence.
5. **a. One-stage, open, perineal anastomotic urethroplasty is preferred.** Posterior urethral reconstruction including excision of the fibrotic segment with distal urethral mobilization and primary anastomosis is associated with the best long-term outcomes after urethral disruption. Incontinence occurs in less than 5% of patients.
6. **a. Retrograde urethrography.** Retrograde urethrography is the most reliable imaging study for urethral evaluation.
7. **e. Immediate primary repair of the left testis.** Immediate primary repair should be attempted in the setting of subtotal injury to an otherwise viable testis. Even extensive testicular injuries often can be safely salvaged, and tunica vaginalis grafts provide better outcomes than do synthetic grafts for complex repair.
8. **b. Spatulated, stented, tension-free, watertight repair of the urethra with absorbable sutures.** Immediate urethral repair with fine absorbable suture over a Foley catheter is associated with superior outcomes after penetrating injury. A proximal bulbar urethral pathologic process in a young man is uniquely amenable to primary anastomotic repair.
9. **c. Retrograde urethrography should be uniformly performed to assess for urethral injury.** Flexible cystoscopy performed at the time of surgical exploration is the simplest and most sensitive means to assess for urethral injury. Urethrography is of low yield in men with no hematuria, no blood at the meatus, and no voiding symptoms; intraoperative flexible cystoscopy is an appropriate alternative method of urethral evaluation.
10. **b. Tunica vaginalis.** Blood fills the space between the visceral and parietal layers of the tunica vaginalis.
11. **b. Is often diagnosed by the presence of intratesticular hypoechoic areas on ultrasonography.** Testicular rupture is often difficult to detect clinically. Ultrasound evaluation usually shows intratesticular heterogeneity as a sentinel finding; detection of a defect of the tunica albuginea is less common.

12. **c. Microscopic dorsal vascular and neural reanastomosis is the best method of repair.** Microvascular reanastomosis of the dorsal neurovascular structures is suggested as the preferred treatment modality whenever possible. Reanastomosis of the corporal arteries is not recommended.
13. **a. Foreskin flap for small distal lesions.** Redundant foreskin provides excellent closure when ample viable tissue exists.
14. **a. Penile fracture must be repaired immediately or there is a decrement in erectile function.** Recent data have shown that a delay in surgery of as long as 7 days has no effect on the outcomes of penile fracture.
15. **e. All of the above.** Antegrade urethral realignment may simplify treatment of the defect, and a large-bore suprapubic catheter placed near the midline will promote subsequent identification of the prostatic apex during delayed reconstruction while preventing tube encrustation or obstruction.

Chapter Review

1. Penile fracture generally occurs at the base of the penis in a ventrolateral location where the tunica albuginea is thinnest.
2. If the location of the penile fracture is evident, a vertical ventral penile incision over the injury may be used thus avoiding a degloving incision. If the location of the injury is uncertain or there is an associated urethral injury, a distal circumcising incision should be made; if this incision is used in an uncircumcised patient, a limited circumcision should be performed before closure to prevent persistent edema of the foreskin.
3. Dog bites of the penis are treated with copious irrigation, debridement, and primary closure. Human bites should be irrigated, debrided, treated with antibiotics, and left open.
4. A fractured testis should be explored and repaired because the salvage rate is higher than when conservative nonoperative therapy is used.

5. Ninety percent of bladder ruptures are associated with pelvic fractures; 10% of pelvic fractures are associated with a bladder rupture.
6. Noncomplicated extraperitoneal bladder ruptures may be treated with urethral catheter drainage alone.
7. The bulbomembranous junction is more vulnerable to injury during pelvic fracture than is the prostatomembranous junction; thus the external sphincter is often intact. In children, urethral disruptions generally occur at the bladder neck. In females, the urethral avulsion usually occurs proximally.
8. In females, urethral disruptions should be primarily repaired and vaginal lacerations should be closed.
9. Initial suprapubic cystostomy is the standard of care for major straddle injuries involving the urethra.
10. When intraperitoneal bladder laceration occurs after blunt trauma, a large laceration of the bladder dome is usually produced that predisposes to urinary ascites and/or peritonitis if it is not repaired promptly.
11. CT cystogram must be performed via retrograde distention of the bladder with a dilute contrast medium. A drainage film is required to complete a plain film cystogram.
12. Flexible cystoscopy performed at the time of surgical exploration is the simplest and most sensitive means to assess for urethral injury.
13. In posterior urethral disruptions, patients may undergo urethral realignment or immediate suprapubic tube placement with planned delayed urethroplasty. Patients undergoing urethral realignment are at high risk for developing stricture recurrence.

134

Special Urologic Considerations in Transgender Individuals

Nicolas Lumen, Anne-Françoise Spinoit, and Piet Hoebeke

Keywords

gender-confirming surgery; metoidioplasty; phalloplasty; transgender

Questions

1. What is the most common urologic complication after phallic reconstruction?
    a. Urethral fistula
    b. Urethral stricture
    c. Stress incontinence
    d. Urge incontinence
    e. Urinary tract infection
2. What is the most frequent site of urethral stricture in transmen?
    a. The anastomotic region between the fixed and phallic parts of the urethra
    b. The fixed part of urethra
    c. The phallic part of urethra
    d. The bladder neck
    e. The neomeatus
3. Rectal injury as a consequence of vaginectomy occurs in approximately what percent of procedures?
    a. 0.09%

b. 0.9%
c. 9%
d. 19%
e. 25%

4. The scrotum in transmen is made of the:
    a. labia majora
    b. labia minora
    c. anterior vaginal wall
    d. perineal skin
    e. labia majora and minora
5. Urologic preoperative assessment in transmen should absolutely include:
    a. uroflowmetry.
    b. ultrasound of the urinary tract.
    c. the patient's urologic history.
    d. all of the above.
    e. only c.
6. Transmen are usually difficult to catheterize transurethrally after surgery because of:
    a. the sharp curve between the female native urethra and the pars fixa of the neourethra.
    b. the presence of a vaginal recessus close to the native female urethra.
    c. hairy skin in the phallic urethra.
    d. the presence of the clitoris dorsally in the perineal urethra.
    e. the difference in elasticity between the phallic and the perineal urethra.
7. Overall, the complication rate in gender-confirming surgery for transmen is:
    a. 0.4%.
    b. 4%.
    c. 40%.
    d. 80%.
    e. 25%.
8. An erectile and testicular prosthesis is implanted:
    a. at the time of phalloplasty.

b. 1 year after phalloplasty.
c. 3 months after phalloplasty.
d. 2 years after phalloplasty.
e. first a testicular and 3 months later an erectile prosthesis.

9. When gender identity and sex do not match in a person, it is called:
   a. transsexualism.
   b. sex reversal.
   c. gender dysphoria.
   d. gender nonconformity.
   e. homosexuality.
10. In transwomen, ultrasound examination of the prostate is best done:
   a. transrectally.
   b. transabominally.
   c. transvaginally.
   d. perineally.
   e. by CT scan only.
11. Which statement is correct about gender identity disorder?
   a. It can be diagnosed at pediatric age.
   b. Is more frequent in biological females than males.
   c. Is based on hormonal imbalance.
   d. Always calls for psychotherapy before any hormonal or surgical treatment.
   e. Is not mentioned in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5).

Answers

1. **a. Urethral fistula.**Urethral fistula is the most common urologic complication after phallic reconstruction.
2. **a. The anastomotic region between the fixed and phallic parts of the urethra.**Multiple retrospective studies in large cohorts of patients show that the anastomotic region between the fixed and phallic parts of the urethra is the most frequent site of urethral stricture in transmen.

3. **b. 0.9%.**Multiple retrospective studies in large cohorts of patients show that rectal injury as a consequence of vaginectomy occurs in approximately 0.9% of patients.
4. **a. Labia majora.**The labia majora are the embryologic counterpart of the scrotum, so using them for scrotal reconstruction is logical and gives the best aesthetic outcomes.
5. **d. All of the above.**In order to exclude any major urologic problem that could interfere with the reconstructive surgery, all three assessments are necessary.
6. **a. Tthe sharp curve between the female native urethra and the pars fixa of the neourethra.**Transmen are usually difficult to catheterize transurethrally after surgery because of the sharp curve between the female native urethra and the pars fixa of the neourethra.
7. **c. 40%.**Based on studies published in the field, the main complication rate including all parts of the surgery is 40%.
8. **b. 1 year after phalloplasty.**The implantation is done when sensitivity is observed in the phallus. As the sensitivity is based on a nerve anastomosis, it takes about a year before this anastomosis becomes functional.
9. **c. Gender dysphoria.**This term is based on the International Classification of Disease and DSM classifications and definitions.
10. **c. Transvaginally.**The neovagina is reconstructed between the rectum and the prostate; thus this new structure allows easy access of the ultrasound probe to examine the prostate.
11. **a. It can be diagnosed at pediatric age.**Gender identity disorder (GID), which is mentioned in DSM-5, can be diagnosed in childhood. GID is more frequent in males and is not based on hormonal imbalance. Psychotherapy as treatment has bad results and is no longer recommended in the standards of care.

Chapter Review

1. The urologist has an important role in the treatment and follow-up of transgender individuals.

2. Urinary incontinence is the most frequent complication in transwomen.
3. Be aware of prostate pathology in transwomen.
4. Discuss whether or not to perform vaginectomy in transmen.
5. Urethral fistulas and strictures are frequent complications in transmen and most often occur at the anastomosis between the fixed and phallic part of the urethra.
6. Metoidioplasty is an alternative to phalloplasty in transmen who refuse to accept the complications of phalloplasty.
7. Implantation of an erectile prosthesis in transmen is associated with a higher risk of complications compared to native males.

135

Tumors of the Bladder

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Keywords

bladder cancer; bladder tumors

Questions

1. The average age of patients with bladder cancer in the United States is:
    a. 65.
    b. 69.
    c. 73.
    d. 76.
    e. 78.
2. Inverted papillomas are:
    a. a benign tumor of the bladder.
    b. a precursor to low-grade papillary cancer.
    c. chemotherapy resistant.
    d. an invasive tumor.
    e. best treated with antibiotics.
3. The incidence rate of urothelial cancer:
    a. has been decreasing recently because of less smoking.
    b. is higher in women than in men.
    c. is highest in developed countries.
    d. peaks in the fifth decade of life.

e. is higher in Asia than Europe.

4. The mortality rate of urothelial cancer:
    a. is primarily related to lack of health care access.
    b. has been decreasing since 1990.
    c. is highest in underdeveloped countries.
    d. is proportionally higher in women than in men.
    e. is proportionally higher in white men than in African American men.
5. What is the risk of a white male developing urothelial cancer in his lifetime?
    a. Less than 5%
    b. 20%
    c. 40%
    d. 60%
    e. 80%
6. The most common histologic bladder cancer cell type is:
    a. squamous.
    b. adeno.
    c. urothelial.
    d. small cell.
    e. leiomyosarcoma.
7. The mortality rate from bladder cancer is highest in:
    a. the United States.
    b. England.
    c. South America.
    d. China.
    e. Egypt.
8. Recent evidence suggests that physician practice may be related to bladder cancer deaths in the elderly. What percentage of deaths could be avoided?
    a. Less than 5%
    b. 30%
    c. 50%
    d. 70%
    e. 90%

9. Which gene is most commonly mutated in high-grade muscle invasive urothelial cancer?
    a. *Cyclin A*
    b. *TP53*
    c. *FGFR-3*
    d. *HRAS*
    e. *PTEN*
10. Which gene is most commonly mutated in carcinoma in situ (CIS)?
    a. *PI3K*
    b. *RB*
    c. *FGFR-3*
    d. *HRAS*
    e. *CD-44*
11. Which gene is most commonly mutated low-grade papillary urothelial carcinoma (LgTa)?
    a. *PTEN*
    b. *RB*
    c. *FGFR-3*
    d. *HRAS*
    e. *CD-44*
12. The chemotherapy proven to cause urothelial cancer is:
    a. doxorubicin.
    b. bleomycin.
    c. ifosfamide.
    d. etoposide.
    e. cyclophosphamide.
13. The increased risk of developing bladder cancer for a man who has a sister with bladder cancer is:
    a. twofold.
    b. 10-fold.
    c. 20-fold.
    d. 40-fold.
    e. 60-fold.
14. The risk of a family member developing bladder cancer if a first-degree relative has the disease is:
    a. related to secondhand smoke.

b. higher in men.
c. higher in smokers.
d. related to inheritance of low-penetrance genes.
e. most common in high-grade cancer.

15. The percent of patients presenting with non–muscle-invasive disease is:
    a. less than 5%.
    b. 20%.
    c. 40%.
    d. 60%.
    e. 80%.
16. A 30-year-old man has gross hematuria, and cystoscopy reveals a papillary tumor. Transurethral resection of the tumor reveals a noninvasive 2-cm papillary low-malignant-potential urothelial tumor. Muscle is present in the resected specimen. All of the tumor is resected. The best treatment is:
    a. intravesical bacille Calmette-Guérin (BCG).
    b. repeat cystoscopy with random bladder biopsies.
    c. radical cystectomy because of the patient's young age.
    d. immediate mitomycin C intravesical therapy.
    e. observation.
17. The external agent most implicated in causing urothelial cancer is:
    a. β-naphthylamine.
    b. 4-aminobiphenyl.
    c. perchloroethylene.
    d. trichloroethylene.
    e. 4,4′-methylene bis(2-methylaniline).
18. If a woman stops smoking for 10 years after 30 pack-years of smoking, her risk of developing bladder cancer:
    a. is the same as if she still smoked.
    b. is the same as if she never smoked.
    c. is unrelated to the intensity of smoking.
    d. is very low because of her gender.
    e. gradually decreases with time.
19. A man exposed to high doses of radiation (more than 500 mSv):

a. has the same risk of urothelial cancer formation as a nuclear plant worker.
b. will likely develop urothelial cancer within 5 years.
c. is more likely to develop urothelial cancer if he is younger than 20 years.
d. is two times more likely to develop urothelial cancer.
e. should be quarantined for 3 months.

20. One of the main changes between the AJCC 7th edition and AJCC 8th edition for bladder cancer is:
    a. there should be two grades of non-muscle-invasive bladder cancer.
    b. multiple positive pelvic lymph nodes is considered stage 3 disease.
    c. perivesical fat involvement by tumors is stage T3.
    d. CIS can be low or high grade.
    e. Ta grade 1 tumors should be considered cancerous.
21. Genetic abnormalities associated with low-malignant potential Ta tumors include:
    a. fibroblast growth factor receptor-3 (*FGFR-3*).
    b. TP53.
    c. retinoblastoma (RB) gene.
    d. PTEN.
    e. loss of chromosome 17.
22. A 40-year-old man has a T1 high-grade urothelial cancer on initial presentation. Muscle was present in the biopsy specimen. The next treatment is:
    a. BCG.
    b. repeat transurethral resection of a bladder tumor (TURBT).
    c. radical cystectomy.
    d. immediate mitomycin C instillation.
    e. neoadjuvant chemotherapy followed by radical cystectomy.
23. A 73-year-old man with a history of Ta bladder cancer is found to have a 0.5-cm papillary lesion in the prostatic urethra and undergoes extensive transurethral resection of the prostate, revealing high-grade noninvasive disease of the prostatic urethra without ductal or stromal involvement. The next best step is:

a. perioperative mitomycin C.
b. surveillance cystoscopy every 3 months.
c. mitomycin C therapy.
d. induction of and maintenance with BCG therapy.
e. radical cystectomy.

24. A 62-year-old man undergoes a transurethral biopsy of a bladder tumor at the dome. Final pathology reveals muscle-invasive urothelial and small cell carcinoma. Metastatic workup is negative. The next step is:
    a. intravesical gemcitabine therapy.
    b. partial cystectomy.
    c. radical cystoprostatectomy.
    d. external beam radiotherapy.
    e. chemoradiation therapy.
25. When cisplatin-based chemotherapy is used, which of the following genetic mutations is associated with the worst prognosis?
    a. *FGFR-3* mutations
    b. *PTEN*
    c. *TP53*
    d. *RB*
    e. *PTEN, TP53,* and *RB*
26. When cisplatin-based neoadjuvant chemotherapy is used, which of the following genetic mutations is associated with an improved response?
    a. *FGFR-3* mutations
    b. *ERCC2*
    c. *TP53*
    d. *RB*
    e. *ERCC2, FANCC,* and *ATM*
27. Tumor suppressor genes are activated by:
    a. gene amplification.
    b. translocation.
    c. point mutations.
    d. DNA methylation.
    e. microsatellite instability.

28. The risk of urologic malignancy in a man with recurrent gross hematuria, but who had a previous negative evaluation, is:
    a. less than 5%.
    b. 20%.
    c. 40%.
    d. 60%.
    e. 80%.
29. Which of the following is not a high-risk factor in urothelial cancer formation in patients with microscopic hematuria?
    a. Age younger than 40 years
    b. Smoking
    c. History of pelvic radiation
    d. Urinary tract infections
    e. Previous urologic surgery
30. Commercially available fluorescence in situ hybridization kits test for abnormalities in which of the following chromosomes?
    a. 3, 7, 9, 17
    b. 2, 5, 8
    c. 4, 6, 9
    d. 1, 10, 12
    e. 13, 14, 16
31. Microsatellite analysis:
    a. detects telomeric repeats.
    b. amplifies DNA repeats in the genome.
    c. evaluates abnormalities on chromosome 9.
    d. detects DNA methylation.
    e. identifies hereditary urothelial cancer.
32. Smoking is responsible for what percent of bladder cancer in males?
    a. 5%
    b. 20%
    c. 40%
    d. 60%
    e. 80%
33. Which of the following is not sensitive to cisplatin chemotherapy?
    a. High-grade urothelial cancer

b. Micropapillary cancer
c. Squamous cell cancer
d. Adenocarcinoma
e. Small cell cancer

34. Nested variant of urothelial cancer can be confused with:
    a. cystitis cystica.
    b. micropapillary cancer.
    c. squamous cell cancer.
    d. small cell cancer.
    e. high-grade urothelial cancer.
35. The most common sarcoma involving the bladder is:
    a. angiosarcoma.
    b. chondrosarcoma.
    c. leiomyosarcoma.
    d. rhabdomyosarcoma.
    e. osteosarcoma.
36. Signet ring cell cancers:
    a. have a good prognosis.
    b. are sensitive to doxorubicin chemotherapy.
    c. usually present in advanced stage.
    d. are responsive to radiation therapy.
    e. are low-grade at initial presentation.
37. The risk of bladder cancer formation in a spinal cord–injured patient is:
    a. less than 5%.
    b. 20%.
    c. 40%.
    d. 60%.
    e. 80%.
38. For patients undergoing radical cystectomy for urothelial cancer, the risk of identifying prostatic urethral disease is:
    a. less than 5%.
    b. 20%.
    c. 40%.
    d. 60%.
    e. 80%.

39. Which of the following is NOT a risk factor for prostatic urethral cancer?
    a. Previous intravesical therapy
    b. CIS of the trigone
    c. CIS of the distal ureters
    d. Low-grade urothelial cancer
    e. Recurrent bladder tumors

Pathology

1. A 70-year-old man has microscopic hematuria. Cytology and computed tomography (CT) scan are negative. Cystoscopy reveals a raised 3-mm lesion on the trigone. The lesion is biopsied and is depicted in Fig. 135.1A and B, and is reported as adenocarcinoma. The next step in management is:
    a. review the pathology slides with the pathologist.
    b. cystectomy.
    c. intravesical chemotherapy.
    d. bone scan.
    e. ask for special stains from pathology.
2. A 65-year-old man has gross hematuria. He has a history of tuberculosis. Cytology is suspicious, CT scan is normal, and cystoscopy reveals a papillary lesion cephalad to the trigone. The lesion is visually completely resected (Fig. 135.2) and is reported as high-grade invasive urothelial carcinoma. The next step in management is:
    a. intravesical BCG.
    b. immediate intravesical mitomycin C.
    c. repeat transurethral resection of bladder at the previous biopsied site.
    d. a cystectomy.
    e. ask the pathologist if there is muscularis propria in the specimen.

Imaging

1. See Fig. 135.3. The depicted findings have an association with:
    a. bladder carcinoma.
    b. previous trauma.
    c. recurrent urinary tract infections.
    d. urolithiasis.
    e. ureteral spasm.

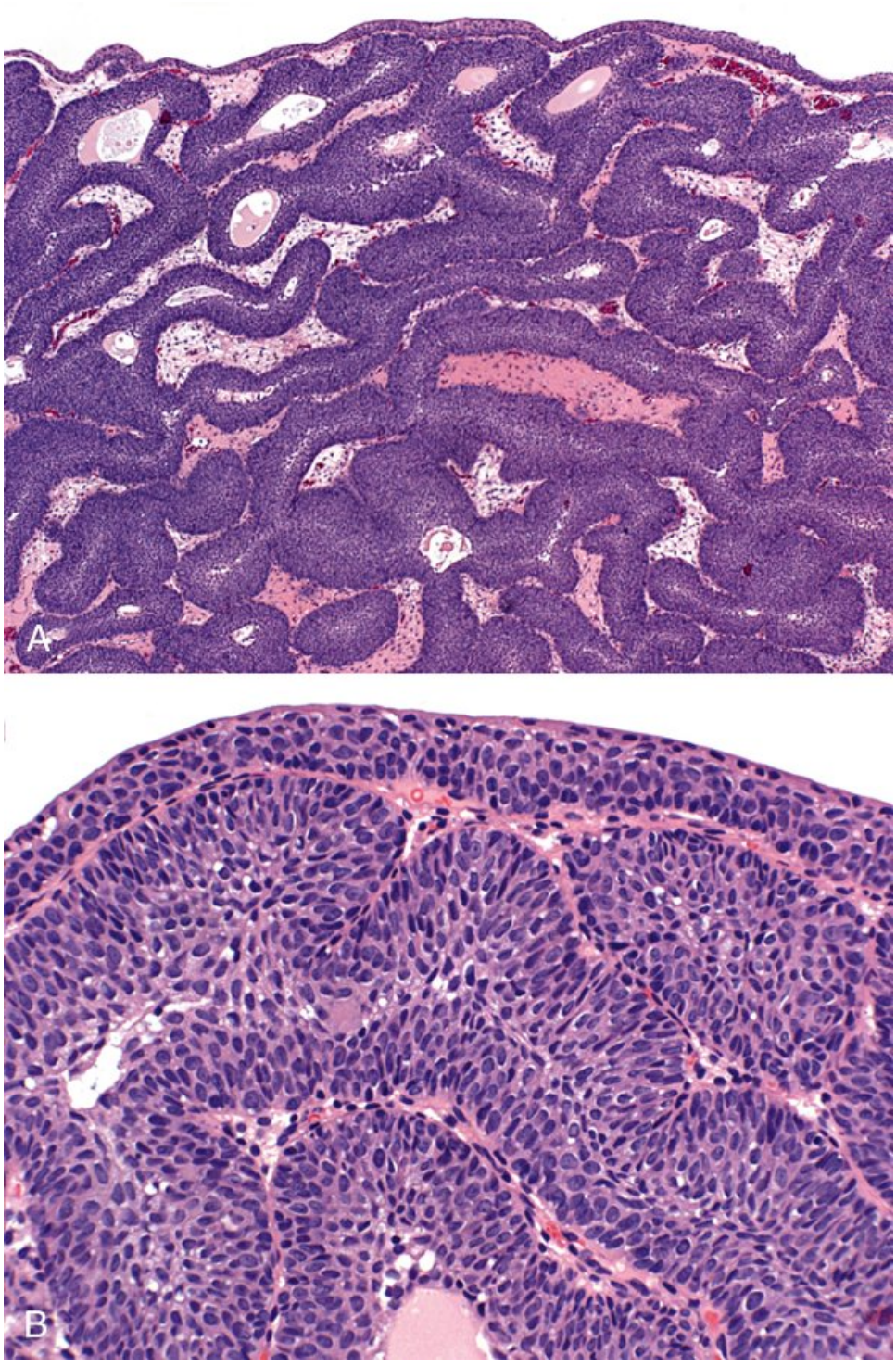


FIG. 135.1
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Mosby, 2008, Edinburgh.

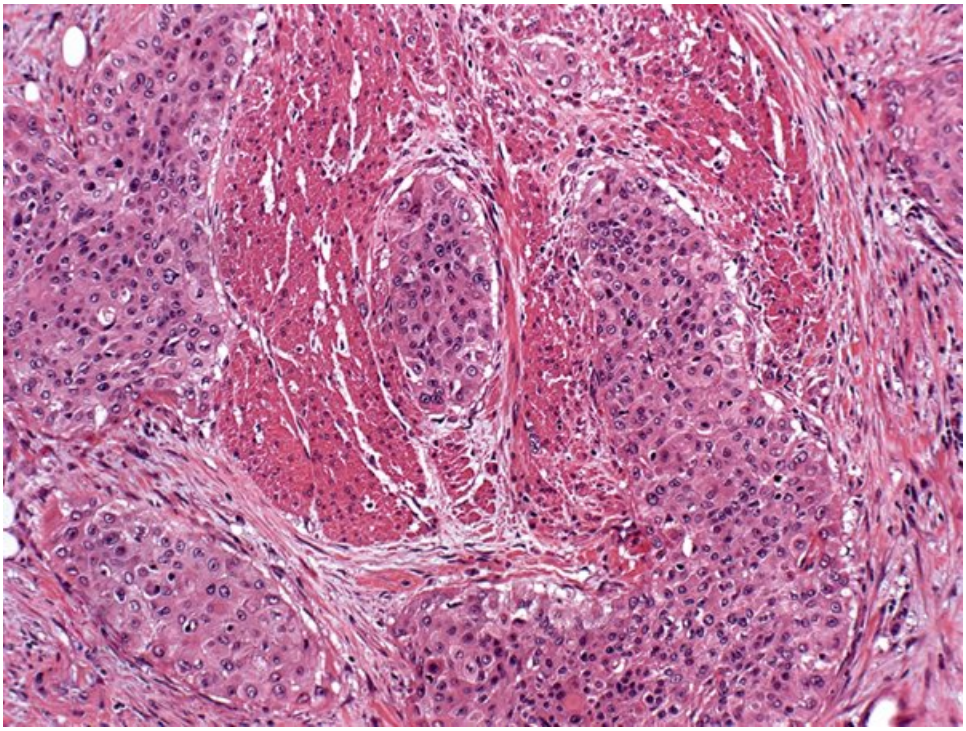


FIG. 135.2
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Mosby, 2008, Edinburgh.

2. Fig. 135.4A is a delayed contrast-enhanced image through the pelvic ureters, and Fig. 135.4B is an early contrast-enhanced image through the bladder. The next step in management is:
    a. shockwave lithotripsy.
    b. percutaneous nephrostolithotomy.
    c. cystoscopy.
    d. cystoscopy with ureteroscopy.
    e. follow-up with imaging in 6 months.

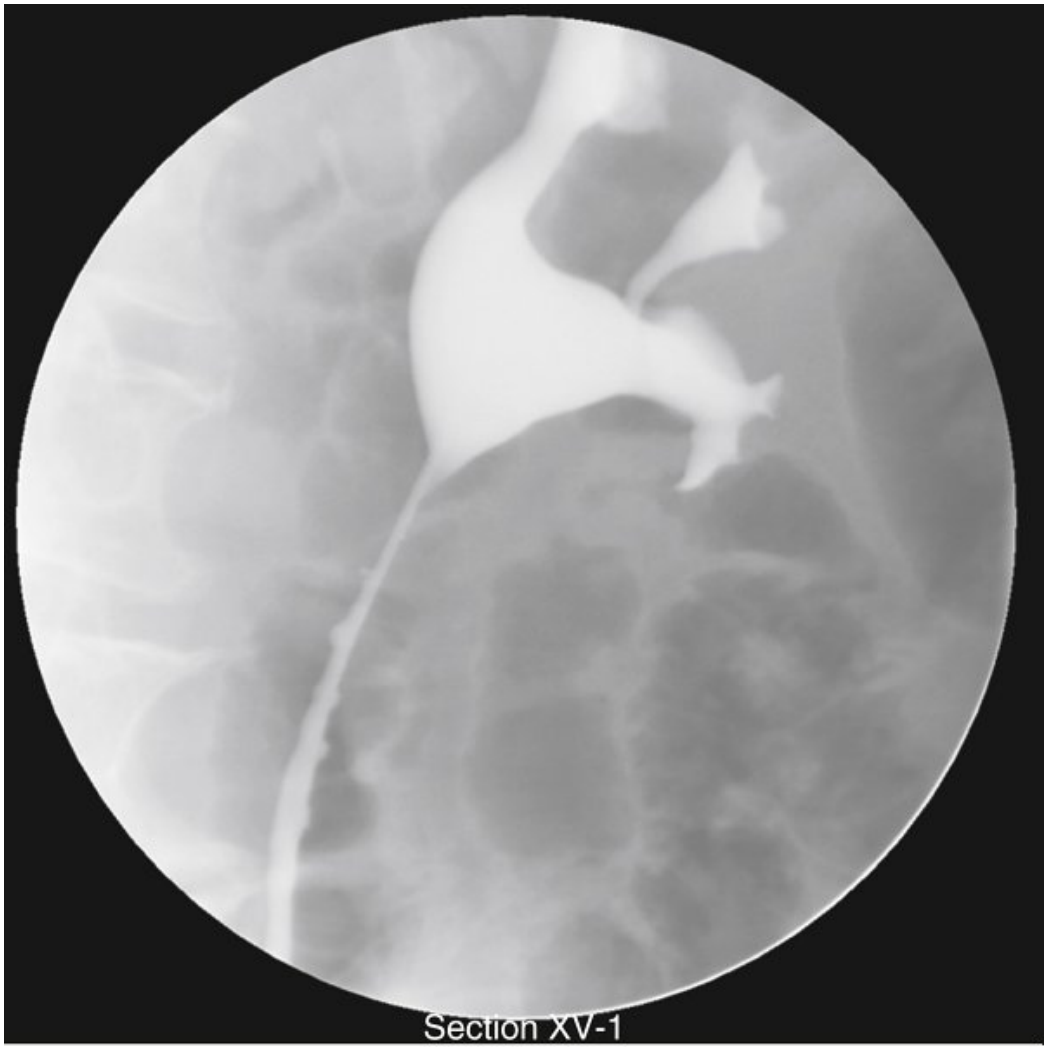


FIG. 135.3


A
Delayed


B
Early

FIG. 135.4

Answers

1. **c. 73.** In the United States, the mean age of diagnosis is 73. [a]

2. **a. A benign tumor of the bladder.** When diagnosed according to strictly defined criteria (e.g., lack of cytologic atypia), inverted papillomas behave in a benign fashion with only a 1% incidence of tumor recurrence (Sung et al., 2006; Kilciler et al., 2008). Occasionally, inverted papillomas are present with coexistent urothelial cancer elsewhere in the urinary system, occurring more commonly in the upper tract than the bladder (Asano et al., 2003). The use of fluorescence in situ hybridization to evaluate chromosomal changes can distinguish between an inverted papilloma and a urothelial cancer with an inverted growth pattern (Jones et al., 2007).
3. **c. Is highest in developed countries.** Sixty-three percent of all bladder cancer cases occur in developed countries, with 55% occurring in North America and Europe. There is a geographic difference in bladder cancer incidence rates across the world, with the highest rates in Southern and Eastern Europe, parts of Africa, Middle East, and North America, and the lowest in Asia and underdeveloped areas in Africa (Ferlay et al., 2007). **The incidence of urothelial cancer peaks in the seventh decade of life.**
4. **b. Has been decreasing since 1990.** The mortality rate of urothelial cancer has decreased by 5% since 1990, primarily because of smoking cessation, changes in environmental carcinogens, and healthier lifestyles (Jemal et al., 2008).
5. **a. Less than 5%.** A white male has a 3.7% chance of developing urothelial cancer in his lifetime, which is roughly 3 times the probability in white females or African American males and more than 4.5 times the probability of an African American female (Hayat et al., 2007; Jemal et al, 2008).

6. **c. Urothelial. Histologically, 90% of bladder cancers are of urothelial origin, 5% squamous cell, and less than 2% adenocarcinoma or other variants** (Lopez-Beltran, 2008). Urothelial carcinoma is the most common malignancy of the urinary tract and is the second most common cause of death among genitourinary tumors.
7. **e. Egypt.** The mortality rate from bladder cancer in Egypt is three times higher than in Europe and eight times higher than in North America because squamous cell carcinoma is highly prevalent in Egypt (Parekh et al., 2002).
8. **b. 30%.** Mortality from bladder cancer is highest in elderly persons, particularly those past the age of 80, accounting for the third most common cause of cancer deaths in men over the age of 80 (Jemal et al., 2008). Whether this increase in mortality rate is related to tumor biology or changes in physician practice with the elderly is unclear. Recent evidence suggests that physician practice may be related to bladder cancer deaths in the elderly (Morris et al., 2009). These authors estimated that 31% of all bladder cancer deaths were avoidable, more commonly in noninvasive than invasive disease.
9. **b. *TP53*. High-malignant potential, non–muscle-invasive bladder cancer is more likely associated with deletions of tumor suppressor genes such as *TP53* and *RB*** (Chatterjee et al., 2004a; George et al, 2007; Sanchez-Carbayo et al, 2007).
10. **b. *RB*. All CIS is high grade by definition. The genetic abnormalities associated with CIS include alterations to the *RB*, *TP53,* and *PTEN* genes** (Cordon-Cardo et al., 2000; Lopez-Beltran et al, 2002; Cordon-Cardo, 2008).
11. **c. *FGFR-3*.** Low-grade urothelial carcinoma is typically papillary in nature with a fibrovascular stalk and frequent papillary branching with increased cellular size, some nuclear atypia, and occasional mitotic figures. These tumors almost universally display alterations of genes in chromosome 9 and frequent mutations in FGFR3, PI3K or, alternatively, Ras genes.
12. **e. Cyclophosphamide. The only chemotherapeutic agent that has been proven to cause bladder cancer is cyclophosphamide** (Travis et al., 1995; Nilsson and Ullen, 2008). The risk of bladder

cancer formation is linearly related to the duration and intensity of cyclophosphamide treatment, supporting a causative role. Phosphoramide mustard is the primary mutagenic metabolite that causes bladder cancer in patients exposed to cyclophosphamide.

13. **a. Twofold. First-degree relatives of patients with bladder cancer have a twofold increased risk of developing urothelial cancer themselves, but high-risk urothelial cancer families are relatively rare** (Aben et al., 2002; Murta-Nascimento et al., 2007; Kiemeney, 2008).
14. **d. Related to inheritance of low-penetrance genes.** The hereditary risk seems to be higher for women and nonsmokers, but it is not related to secondhand exposure to smoking in families. Most likely, there are a variety of low-penetrance genes that can be inherited to make a person more susceptible to carcinogenic exposure, thus increasing the risk of bladder cancer formation.
15. **e. 80%. At initial presentation, 80% of urothelial tumors are non–muscle-invasive.** There are multiple growth patterns of urothelial cancer, including flat carcinoma in situ (CIS), papillary tumors that can be low or high grade, and sessile tumors with a solid growth pattern. Non–muscle-invasive cancers can be very large because of lack of the genetic alterations required for invasion.
16. **d. Immediate mitomycin C intravesical therapy. PUNLMP is a papillary growth with minimal cytological atypia that is more than seven cells thick and is generally solitary and located on the trigone** (Holmang et al., 2001; Sauter et al., 2004). PUNLMP is composed of thin papillary stalks where the polarity of the cells is maintained and the nuclei are minimally enlarged. PUNLMP has a low proliferation rate and is not associated with invasion or metastases. Tumor recurrence is common, and thus perioperative treatment with mitomycin C is warranted.
17. **a. β-naphthylamine.** One of the first and most common chemical agents implicated in the formation of bladder cancer in dye and rubber workers is β-naphthylamine (Case and Hosker, 1954). Activation of aromatic amines allows DNA binding by enzymes that are selectively expressed in the population, making some

subjects more susceptible to cancer formation, as described earlier related to the *NAT-2* and the *GSTM1* polymorphisms.

18. **e. Gradually decreases with time. Smoking cessation does make a difference in urothelial cancer formation.** Smokers who have stopped for 1 to 3 years have a 2.6 relative risk, and those who have stopped for more than 15 years have a 1.1 relative risk of bladder cancer formation (Wynder and Goldsmith, 1977; Smoke IAfRoCT, 2004).
19. **d. Is two times more likely to develop urothelial cancer.** There is a significant increased risk of dying from any cancer if a person is exposed to greater than 50 mSv. The relative risk of urothelial cancer formation is 1.63 in men and 1.74 in women. **Interestingly, urothelial cancer formation after radiation is not age related, but the latency period is 15 to 30 years.** However, there is no association with low-dose or industrial exposure of radiation therapy and bladder cancer formation. Importantly, urologic technicians and nuclear radiation workers do not have an increased risk of urothelial cancer formation.
20. **b. Multiple positive pelvic lymph nodes is considered stage 3 disease.** Broadly speaking, non–muscle-invasive bladder cancer is composed of stage 0 (noninvasive) and stage 1 (invasion into subepithelial connective tissue), muscle-invasive organ-confined bladder cancer is of stage 2, muscle-invasive locally advanced bladder cancer is stage 3, and metastatic disease is stage 4. Whereas prior AJCC editions viewed positive lymph nodes as stage 4 regardless of the quantity or location, the 2017 AJCC staging update now views nodal disease in the absence of systemic metastases as stage 3.
21. **a. Fibroblast growth factor receptor-3 (*FGFR-3*). Genetic abnormalities associated with low-grade cancer include deletion of 9q and alterations of *FGFR-3*, *HRAS*, and *PI3K*** (Holmang et al, 2001; Cordon-Cardo, 2008). Low-grade carcinomas are immunoreactive for cytokeratin-20 and CD-44. **The *TP53,* retinoblastoma *(RB),* and *PTEN* genes and loss of chromosome 17 are all associated with high-grade cancer.**

22. **b. Repeat transurethral resection of a bladder tumor (TURBT). Because of this understaging, the American Urological Association (AUA) guidelines call for a repeat transurethral resection in patients with T1 tumors to assess for muscle-invasive disease even if muscle was present in the specimen** (Hall et al, 2007).
23. **d. Induction of and maintenance with BCG therapy.** For patients with noninvasive prostatic urethral cancer, transurethral resection of the prostate with BCG therapy is appropriate (Palou et al, 2007). For patients with prostatic ductal disease, a complete TURP is warranted, plus BCG therapy. Although a radical cystectomy could be performed, a more conservative organ-sparing treatment is recommended.
24. **e. Chemoradiation therapy.** Small cell carcinoma of the bladder should be considered and treated as metastatic disease, even if there is no radiologic evidence of disease outside the bladder. Small cell carcinoma of the bladder accounts for much less than 1% of all primary bladder tumors. **In general, small cell carcinoma of the bladder is very chemosensitive, and the primary mode of therapy is chemoradiation therapy or chemotherapy followed by cystectomy.**
25. **e. *PTEN, TP53,* and *RB*. Overall genetic instability is the hallmark of invasive urothelial cancer, but, specifically, alterations of *TP53, RB,* and *PTEN* carry a very poor prognosis** (Chatterjee et al, 2004a). *FGFR-3* mutations are associated with noninvasive bladder cancer.
26. **e. *ERCC2, FANCC,* and *ATM* .** The recent application of genomics to large cohorts of MIBCs has produced major breakthroughs in disease heterogeneity with obvious implications for clinical management. Whole transcriptome studies identified basal and luminal molecular subtypes (Choi et al., 2014; TCGAR, 2014; Damrauer et al., 2014), and patients with basal tumors appeared to derive the most benefit from neoadjuvant chemotherapy (NAC) (Seiler et al 2017; McConkey et al., 2018). In parallel, two other groups demonstrated that inactivating mutations in DNA damage repair genes (including ERCC2, FANCC, ATM)

were associated with response to NAC (Van Allen et al., 2014; Serebriiskii et al., 2015). These findings are now being prospectively examined within the context of the Southwest Oncology Group's (SWOG's) completed S1314 trial. In addition, two groups have designed prospective clinical trials to test whether DDR mutations can be used to guide bladder preservation in patients treated with NAC, and several groups are considering clinical trials to select MIBC patients for NAC based on basal molecular subtype membership.

27. **c. Point mutations.** Tumor suppressor genes are mainly activated by allelic deletion of one allele followed by point mutations of the remaining allele. Tumor suppressor genes are recessive or have a negative effect, resulting in unregulated cellular growth. Proto-oncogenes are generally activated by point mutations in the genetic code, gene amplification, and gene translocation. The activated proto-oncogenes become oncogenes that can cause cancer, and this is considered a positive or dominant growth effect (Lengauer et al, 1998; Wolff et al, 2005; Cordon-Cardo, 2008).
28. **a. Less than 5%. Gross, painless hematuria is the primary symptom in 85% of patients with a newly diagnosed bladder tumor** (Khadra et al, 2000; Alishahi et al, 2002; Edwards et al, 2006). The gross hematuria is usually intermittent and can be related to Valsalva maneuvers; therefore any episode of gross hematuria should be evaluated even if subsequent urinalysis is negative. Fifty percent of patients with gross hematuria will have a demonstrable cause, 20% will have a urological malignancy, and 12% will have a bladder tumor (Khadra et al, 2000). **The risk of malignancy in patients with recurrent gross or microscopic hematuria that had a full, negative evaluation is near zero within the first 6 years** (Khadra et al, 2000).
29. **a. Age younger than 40 years.** The guidelines recommend consideration for reevaluation of low-risk individuals with microscopic hematuria, but repeat evaluation every 6 months with a urinalysis, cytology, and blood pressure (to detect renal disease) is recommended for high-risk patients. Age younger than 40 years

is the only factor that is not associated with an increased risk of malignancy.

30. **a. 3, 7, 9, 17.** Fluorescence in situ hybridization (FISH) identifies fluorescently labeled DNA probes that bind to intranuclear chromosomes. **The current commercially available probes evaluate aneuploidy for chromosomes 3, 7, 17, and homozygous loss of 9p 21** (Zwarthoff, 2008). The median sensitivity and specificity of FISH analysis is 79% and 70%, respectively (van Rhijn et al, 2005).
31. **b. Amplifies DNA repeats in the genome.** There are multiple markers available to identify short DNA repeats present throughout the chromosomes that are lost in some tumor cells. Microsatellite analysis amplifies these repeats in the genome that are highly polymorphic, and PCR amplification can detect tumor-associated loss of heterozygosity by comparing the peak ratio of the two alleles in tumor DNA in a urine sample with that ratio in a blood sample from the same individual (Steiner et al, 1997; Wang et al, 1997). The sensitivity and specificity of microsatellite analysis for the detection of urothelial carcinoma range from 72% to 97% and 80% to 100%, respectively (Steiner et al, 1997; Wang et al, 1997). Microsatellite analysis evaluates abnormalities on all chromosomes.
32. **c. 40%.** Smoking is responsible for 30% to 50% of all bladder cancers in males, and smokers have a twofold to sixfold greater risk for bladder cancer (Brennan et al, 2000; Boffetta, 2008). Smoking cessation will decrease the risk of eventual urothelial cancer formation in a linear fashion. After 15 years of not smoking, the risk of cancer formation is the same as for a person who never smoked (Smoke IAfRoCT, 2004). The strong influence of smoking in bladder cancer formation prevents accurate determination of other, less significant dietary, micronutrient, or lifestyle changes that may alter bladder cancer formation.
33. **b. Micropapillary cancer.** The most effective treatment for all stages of micropapillary urothelial carcinoma is surgical resection. Treatment with transurethral resection and BCG therapy is ineffective unless the tumor is completely resected (Kamat et al,

2007). **Neoadjuvant chemotherapy does not appear effective in micropapillary urothelial carcinoma, similar to ovarian cancer** (Bristow et al, 2002; Kamat et al, 2007). Neoadjuvant chemotherapy may actually worsen survival by delaying therapy when compared with immediate cystectomy. Cisplatin is effective against urothelial cancer and the associated variants of squamous cell, adenocarcinoma, and small cell cancer.

34. **a. Cystitis cystica.** The nested variant of urothelial cancer is a rare but aggressive cancer that has a male-to-female ratio of 6:1 and can be confused with benign lesions, such as Von Brunn nests that are in the lamina propria, cystitis cystica, and inverted papillomas (Holmang and Johansson, 2001). There is little nuclear atypia in nested variant urothelial carcinoma, but the tumor cells will often contain areas with large nuclei and mitotic figures. **The mortality rate from nested variant urothelial carcinoma, despite aggressive therapy, is significant, with 70% dying of their disease within 3 years** (Paik and Park, 1996).
35. **c. Leiomyosarcoma. Leiomyosarcoma is the most common histologic subtype, followed by rhabdomyosarcoma and then, rarely, angiosarcomas, osteosarcomas, and carcinosarcomas.** The male-to-female ratio is 2:1, and the average age at presentation is in the sixth decade of life. There are no clear agents that cause bladder sarcomas, although there is an association with pelvic radiation and systemic chemotherapy for other malignancies (Spiess et al, 2007). Importantly, bladder sarcomas are not smoking related.
36. **c. Usually present in advanced stage.** Primary signet ring cell carcinoma of the bladder is extremely rare, making up less than 1% of all epithelial bladder neoplasms (Morelli et al, 2006). **Signet ring cell carcinoma can be of urachal origin and directly extend into the bladder. These tumors generally present as high-grade, high-stage tumors and have a uniformly poor prognosis.** The primary treatment is radical cystectomy; however, in the majority of cases there are regional or distant metastases at the time of presentation, and the mean survival time is less than 20 months (Torenbeek et al, 1996). There are reports of elevated

carcinoembryonic antigen (CEA) in patients with signet ring cell carcinoma. The prognostic significance of this elevated serum marker is unclear (Morelli et al, 2006). Understaging is very common in signet ring cell carcinoma, with peritoneal studding common at the time of surgical exploration.

37. **a. Less than 5%.** Spinal cord–injured patients are at risk for developing squamous cell carcinoma, most likely due to chronic catheter irritation and infection. Older studies have suggested a 2.5% to 10% incidence of squamous cell carcinoma in the spinal cord–injured population, with a mean delay of 17 years after the spinal cord injury (Kaufman et al., 1977). More recent analysis of the association of spinal cord injury and bladder cancer formation has shown a remarkably lower risk of bladder cancer formation of 0.38%, most likely because of better catheter care (Bickel et al., 1991). This supports the concept that chronic infection and foreign bodies can lead to bladder cancer formation.
38. **c. 40%.** Prostatic urethral cancer is associated with urothelial cancer of the bladder in 90% of cases, primarily CIS, and most will have multifocal bladder tumors. However, the incidence of prostatic urethral disease in patients with primary urothelial cancer is only 3% (Rikken et al., 1987; Millan-Rodriguez et al., 2000). **Secondary prostatic urethral involvement in patients with a history of urothelial cancer is approximately 15% at 5 years and 30% at 15 years, almost uniformly associated with extensive intravesical therapy** (Herr and Donat, 1999). For patients undergoing radical cystectomy for urothelial cancer, the risk of identifying prostatic urethral disease is 40%.
39. **d. Low-grade urothelial cancer. Risk factors for prostatic urethral involvement are CIS of the trigone, bladder neck, distal ureters, recurrent bladder tumors, and a history of intravesical chemotherapy** (Wood et al., 1989b). Low-grade tumors rarely involve the prostatic urethra.

Pathology

1. **a. Review the pathology slides with the pathologist.** This is a classic inverted papilloma, and the inexperienced pathologist might mistake it for an adenocarcinoma. The location of the lesion would be unusual for adenocarcinoma, particularly in a patient with no risk factors, and should alert the clinician to review the slides with the pathologist.
2. **e. Ask the pathologist if there is muscularis propria in the specimen.** There are clear bundles of muscularis propria in the micrograph making the tumor at least a T2.

Imaging

1. **a. Bladder carcinoma.** Pseudodiverticulosis of the ureter is associated with bladder carcinoma in 30% of cases. This association has led many to recommend that patients with this diagnosis undergo surveillance of their bladder for the development of urothelial neoplasms. The etiology is unknown.
2. **d. Cystoscopy with ureteroscopy.** There are multiple enhancing masses in the fluid-filled urinary bladder on the early image. On the delayed image, the ureters are opacified with contrast, and there is a filling defect seen in the mildly dilated right ureter, suspicious for a synchronous ureteral lesion.

Chapter Review

1. Inverted papillomas are associated with chronic inflammation.
2. Cystitis glandularis may be associated with pelvic lipomatosis.
3. Bladder cancers in adolescents and young adults generally are well differentiated and noninvasive.
4. The intensity and duration of smoking is linearly related to the risk of developing bladder cancer with no plateau; cessation of smoking reduces the risk.
5. There is a clear association between a healthy diet and a decreased risk of urothelial cancer.

6. There is no convincing evidence that alteration in fluid intake, alcohol consumption, ingestion of artificial sweeteners, or analgesic abuse increase the risk of bladder cancer; however, chronic irritation, bacterial infection, and radiation have all been associated with the development of bladder cancer.
7. Eighty percent of the time, low-grade, low-stage urothelial neoplasia (papillary urothelial neoplasia of low malignant potential) is associated with loss of chromosome 9.
8. Low-grade (stage 1), low-stage urothelial neoplasia is called papillary urothelial neoplasia of low malignant potential (PUNLMP); the terms low grade and high grade replace the old system of grades 2 and 3.
9. Prostatic urethral involvement by transitional cell carcinoma without invasion carries a relatively good prognosis; when it invades the prostatic stroma, the prognosis is less good, and when it directly invades the substance of the prostate from the bladder, the prognosis is poor.
10. Low-grade papillary lesions have a 60% recurrence rate but less than a 10% rate of progression to muscularis propria invasion, whereas high-grade lesions, particularly T1, may have a stage progression in as many as 50% of cases. Moreover, high-grade non-muscularis propria invasive tumors have an 80% incidence of recurrence.
11. Angiolymphatic invasion is a poor prognostic sign.
12. In muscularis propria invasive urothelial cancer, alterations in *TP53, RB,* and *PTEN* are poor prognostic indicators.
13. Genetic alterations in low-grade non-muscularis propria invasive disease include alterations in *FGFR-3* and deletions in chromosome 9.
14. Porphyrin-induced fluorescent cystoscopy and narrow-band imaging cystoscopy have been used to increase the sensitivity of cystoscopy.
15. To date, none of the urinary markers are sensitive or specific enough to replace cystoscopy for monitoring bladder cancer.
16. Sarcomas of the bladder, in decreasing order of frequency, include leiomyosarcoma, rhabdomyosarcoma, angiosarcoma, osteosarcoma, and carcinosarcoma.

17. *Schistosoma haematobium* is the causative agent of squamous cell carcinoma in endemic regions.
18. Altered growth patterns, such as micropapillary and nested patterns, carry a poor prognosis.
19. Normal bladder urothelium is less than seven cell layers thick: papillary lesions are greater than seven cell layers thick.
20. The incidence of urothelial cancer peaks in the seventh decade of life.
21. There is some evidence to indicate that BCG plus oral administration of vitamins A, B_6, C, E, and zinc result in a reduced risk of recurrent transitional cell carcinoma.
22. Histologically, 90% of bladder cancers are of urothelial origin, 5% squamous cell, and less than 2% adenocarcinoma or other variants.
23. All CIS is high grade by definition. The genetic abnormalities associated with CIS include alterations to the *RB*, *TP53,* and *PTEN* genes.
24. The only chemotherapeutic agent that has been proven to cause bladder cancer is cyclophosphamide.
25. First-degree relatives of patients with bladder cancer have a twofold increased risk of developing urothelial cancer themselves.
26. Urothelial cancer formation after radiation is not age related; the latency period is 15 to 30 years.
27. A repeat transurethral resection in patients with T1 tumors to assess for muscle-invasive disease should be performed, even if muscle was present in the original specimen.
28. Small cell carcinoma of the bladder is very chemosensitive.
29. The risk of malignancy in patients with recurrent gross or microscopic hematuria who had a full, negative evaluation is near zero within the first 6 years.
30. Neoadjuvant chemotherapy does not appear effective in micropapillary urothelial carcinoma.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

136

Management Strategies for Non–Muscle-Invasive Bladder Cancer (Ta, T1, and CIS)

Joseph Zabell, and Badrinath Konety

Keywords

bacillus Calmette-Guérin; cystoscopy; intravesical therapy; mitomycin C; non–muscle-invasive bladder cancer; urinary markers

Questions

1. Postoperative intravesical chemotherapy (administered in the recovery room) is most appropriate for which of the following patients?
    a. Initial presentation of a solitary 3.0-cm, low-grade–appearing tumor on the posterior bladder wall
    b. Multifocal high-grade Ta lesions with associated urothelial carcinoma in situ (CIS)
    c. 6.5-cm high-grade, broad-based tumor on lateral wall with deep resection
    d. a and b
    e. a, b, and c
2. Which of the following agents is contraindicated for immediate postoperative intravesical chemotherapy (administered at time of transurethral resection of bladder tumor [TURBT])?

a. Thiotepa
b. Bacille Calmette-Guérin (BCG)
c. Mitomycin C (MMC)
d. Epirubicin
e. b and c

3. Potential advantage(s) of tumor markers such as, BTA, stat NMP-22, and UroVysion (FISH) when compared with urinary cytology for monitoring patients with bladder cancer is (are) improved:
   a. sensitivity.
   b. specificity.
   c. positive predictive value.
   d. a and c.
   e. a, b, and c.
4. Progression rates for low-grade Ta tumors range from:
   a. 3% to 10%.
   b. 12% to 17%.
   c. 17% to 25%.
   d. 25% to 35%.
   e. Over 35% and higher
5. General anesthesia can be advantageous compared with spinal anesthesia when a bladder tumor is being resected in which setting?
   a. Large mobile papillary tumor
   b. Tumor in a posterior wall diverticulum
   c. Lateral location at approximately 4 or 8 o'clock
   d. Extensive CIS
   e. Tumor at dome and along anterior bladder wall
6. A healthy 55-year-old man undergoes resection of a 2.0-cm bladder tumor in a posterior wall bladder diverticulum. Pathology demonstrates a pT1G3 bladder tumor with associated areas of CIS. Muscularis mucosa is involved, but there is no definite muscularis propria in the specimen. Optimal management includes
   a. repeat resection to stage the cancer.
   b. intravesical BCG therapy.
   c. partial cystectomy with excision of the diverticulum.
   d. radical cystectomy and urinary diversion.
   e. chemotherapy and radiation therapy.

7. The most important principle to follow when tumor is resected near or overlying a ureteral orifice is which?
    a. A ureteral stent must be placed at time of resection.
    b. In most cases resection is to be avoided.
    c. Coagulating current should not be used in this area.
    d. Resection and fulguration may be done at will—a stent or nephrostomy tube can be placed later.
    e. If hydronephrosis is found, an ultrasound should be obtained preoperatively and a nephrostomy tube placed.
8. A restaging TURBT is indicated in which of the following situations?
    a. pT1 high-grade tumor with no muscularis propria identified
    b. pTa low-grade tumor that is multifocal ($n = 5$), for which resection appeared to be complete but postoperative intravesical therapy was *not* administered
    c. pT1 high-grade tumor with muscularis propria identified and negative
    d. a and c
    e. a, b, and c
9. Intravesical mitomycin C chemotherapy for high-risk superficial bladder cancer:
    a. reduces the risk of progression.
    b. reduces the risk of recurrence.
    c. is preferred over BCG, particularly for CIS.
    d. is virtually free of side effects.
    e. is less expensive than BCG.
10. Relative contraindications to intravesical BCG therapy include which of the following?
    a. cirrhosis
    b. history of tuberculosis (TB)
    c. total incontinence
    d. immunosuppression
    e. all of the above
11. The combination of reduced-dose BCG and interferon alpha for intravesical therapy is:
    a. more effective than BCG alone.

b. more toxic than BCG alone.
c. preferred first-line therapy for patients with multifocal CIS.
d. less effective than BCG alone.
e. a reasonable option for BCG failures after one course of therapy.

12. Common side effects of thiotepa include:
    a. irritative voiding symptoms and fever.
    b. hematuria and irritative voiding symptoms.
    c. bladder contraction and myelosuppression.
    d. irritative voiding symptoms and myelosuppression.
    e. flu-like symptoms and fever.
13. Long-term (15 years) outcomes after intravesical BCG therapy for patients with high-risk non–muscle-invasive bladder cancer include which of the following?
    a. approximately 50% progression rate
    b. approximately 25% alive and with bladder intact
    c. a high incidence of recurrence in extravesical sites (prostatic urothelium and upper tracts)
    d. approximately 33% cancer-related mortality rates
    e. all of the above
14. Understaging for patients with pT1 high-grade bladder cancer is approximately:
    a. 5% to 10%.
    b. 10% to 20%.
    c. 20% to 30%.
    d. 30% to 50%.
    e. 50% to 70%.
15. A patient is diagnosed with a 1.0-cm pTa low-grade bladder cancer. What about imaging of the upper tracts?
    a. Not indicated
    b. Indicated only at diagnosis
    c. At diagnosis and 5 years later
    d. At diagnosis and every other year thereafter
    e. At diagnosis and every year thereafter:
16. For patients with a stage pTa low-grade bladder tumor and a negative cytology, random bladder biopsies:

a. are more likely to be positive in the prostatic fossa than the bladder.
b. must be done in a systematic manner.
c. should include sampling of the muscularis mucosa and preferably also the muscularis propria.
d. are indicated at initial diagnosis and need not be repeated if negative.
e. are not indicated in most cases.

17. The risk of progression to muscle-invasive disease for patients with untreated CIS of the bladder is approximately:
    a. 5% to 15%.
    b. 15% to 25%.
    c. 25% to 35%.
    d. 35% to 45%.
    e. higher than 45%.
18. The most important factor determining the long-term impact of BCG on progression is:
    a. using a full dose with each instillation.
    b. adding interferon.
    c. maintenance therapy.
    d. rolling the patient side to side to ensure that BCG covers the entire urothelium.
    e. washing the toilet with bleach after voiding each dose.
19. According to the American Urological Association (AUA) risk stratification tables from the 2016 AUA/Society of Urologic Oncology (SUO) guideline of non–muscle-invasive bladder cancer (NMIBC), which of the following tumor characteristics would classify as an intermediate risk tumor?
    a. HG T1
    b. Any CIS
    c. Solitary LG Ta >3 cm
    d. PUNLMP
    e. Any variant histology
20. Which of the following represents an appropriate surveillance algorithm for high-risk NMIBC according to the 2016 AUA/SUO guideline?

a. Cystoscopy every 3 to 4 months for 2 years, every 6 months for the subsequent 2 years, and annually for life
b. Cystoscopy every 3 to 6 months for 2 years, then stop
c. Imaging unnecessary unless hematuria is present
d. Upper tract imaging annually for at least 2 years
e. Cystoscopy at 3 months and then again 9 months later

21. A 76-year-old male with a history of urothelial CIS begins induction BCG. Following his second instillation, he notes severe urinary urgency and dysuria as well as a fever to 102°F that is present 2 days after instillation. Appropriate management of this BCG toxicity includes which of the following?
    a. Continue BCG instillations unchanged at next instillation
    b. Start isoniazid and rifampin
    c. Start ciprofloxacin
    d. Consider infectious disease consult
    e. b and d
22. Which of the following genetic mutations is commonly linked to low-grade NMIBC?
    a. Rb
    b. Fibroblast growth factor receptor 3 (FGFR3)
    c. p53
    d. RAS
    e. PTEN
23. In the setting of initial BCG failure, a second course of BCG provides a response rate of approximately:
    a. 5% to 10%
    b. 10% to 20%
    c. 30% to 50%
    d. 60% to 80%
    e. >90%

Answers

1. **a. Initial presentation of a solitary 3.0 cm, low-grade–appearing tumor on the posterior bladder wall. Postoperative intravesical**

chemotherapy should be considered for most cases of new, apparently low-grade non–muscle-invasive bladder cancer because it has been shown to reduce recurrence rates and to improve outcomes for this disease. One major exception is the patient in whom an extensive resection has been performed or whenever there is a possible perforation. In these patients intravesical chemotherapy should be withheld owing to concern about local extravasation and absorption. The benefit of postoperative intravesical chemotherapy is reduced in patients with recurrent or multiple tumors, and there is no clear benefit in patients with high-grade disease.

2. **b. Bacillus Calmette-Guérin (BCG). BCG should never be given in association with known trauma to the urinary tract, such as after transurethral resection of a bladder tumor (TURBT), owing to concern over systemic absorption and sepsis.** All of the other agents have shown efficacy in this setting with a favorable morbidity profile.
3. **a. Sensitivity. Tumor markers such as BTA stat, NMP-22, and UroVysion (FISH) provide increased sensitivity, particularly for low-grade tumors. High specificity is the strength of urinary cytology.** This approaches 90% to 100% in many series and has not been improved on with these other markers. Positive predictive value is highest for urinary cytology because the number of false positives is low.
4. **a. 3% to 10%. Recurrence is common (50%–70%) for patients with low-grade pTa tumors, but progression to higher tumor stage is uncommon, occurring in about 5% to 10% of patients.**
5. **c. Lateral location, at approximately 4 or 8 o'clock. Resection along the lateral bladder wall posterolaterally places one in proximity to the obturator nerve, which can lead to an obturator reflex and increased risk of bladder wall perforation.** In this situation, general anesthesia with complete paralysis is sometimes beneficial to allow the procedure to be performed in a safe and simple manner.
6. **d. Radical cystectomy and neobladder urinary diversion.** This patient should be strongly considered for radical cystectomy. Partial

cystectomy is not a good option because of the presence of CIS, which indicates a high risk of field-effect disease and subsequent recurrence. Deeper biopsies will risk perforation and would be unlikely to influence management. **Understaging is common with tumors in diverticula, and high-grade invasive tumors like this are best managed with radical cystectomy to ensure local disease control and optimize outcomes on a long-term basis.**

7. **c. Coagulating current should not be used in this area.** A stent should be avoided if possible to prevent theoretical risk of reflux of tumor cells into the upper tracts. In most cases this area can be resected, and most ureters will remain unobstructed as long as the orifice is identified and coagulating current is not used in this area. Preoperative placement of a nephrostomy tube is often unnecessary as long as renal function is stable. Many patients with hydronephrosis will have invasive disease and will be undergoing urinary diversion in the near future, and this will relieve the obstruction. Hence, temporary nephrostomy tube placement is usually not required.
8. **d. a and c.** Patients with pT1 tumor for whom the muscularis propria was not identified are understaged about 50% of the time and a repeat resection is clearly indicated. **Repeat resection of patients with pT1G3 tumor with muscularis propria present and negative are found to have residual or invasive disease 30% of the time.** A repeat TURBT is thus indicated in both of these patient populations to accurately stage the tumor and to optimize patient management.
9. **b. Reduces the risk of recurrence. Mitomycin C (MMC) can reduce the risk of recurrence, but there is no convincing evidence that it can reduce progression rates, which is true for all forms of intravesical chemotherapy. Most comparative studies and meta-analyses suggest an advantage to BCG, particularly for CIS.** MMC can lead to local bladder irritation and a number of other side effects and is thus far from risk free.
10. **e. All of the above. BGC is contraindicated in patients with liver disease (isoniazid cannot be given if they develop BCG sepsis), have a personal history of TB, have total incontinence**

(they cannot retain the BCG, so efficacy would be poor). Immunosuppression represents a relative contraindication as the mechanism of action is to stimulate an immune response. Although small series have demonstrated this to be safe, it may be less effective than in immunocompetent patients. Other contraindications include disrupted urothelium, gross hematuria, or active or persistent symptomatic urinary tract infection (UTI).

11. **d. Less effective than BCG alone.** BCG plus interferon alpha has shown activity in BCG failures; this is a viable option for this challenging patient population. However, it is more expensive, and a randomized trial demonstrated that it is less effective than BCG alone in BCG-naive patients. BCG remains the treatment of choice for CIS. BCG plus interferon alpha is well tolerated with a side-effect profile that is better on average than BCG alone, because most of the side effects are related to the BCG and its dose is reduced in this regimen.
12. **d. Irritative voiding symptoms and myelosuppression. Irritative voiding symptoms are reported by 12% to 69% of patients receiving intravesical thiotepa. The low molecular weight of this agent (189 kD) predisposes to systemic absorption and myelosuppression.** These are the two most common side effects of this agent.
13. **e. All of the above. Data about long-term outcomes for patients with high-risk superficial bladder cancer treated with intravesical BCG therapy are derived primarily from the experience at Memorial Sloan Kettering (Cookson et al., 1997).** [a] **In this series, 50% of patient progressed and one-third died of cancer progression. Approximately one-third developed disease in the prostatic fossa or upper tracts and only 27% survived with an intact bladder.** Such data should be considered when patients are being counseled about treatment options for high-risk disease.

14. **d. 30% to 50%.** The risk of understaging a pT1 high-grade bladder tumor is approximately 30%, but it is even higher if there is no muscularis propria in the specimen. Overall, the risk is about 30% to 50% in this high-risk patient population.
15. **a. Not indicated.** The incidence of upper tract tumor associated with pTaG1 bladder cancer is extremely low (0.3%–2.3%), and current consensus is that upper tract imaging is not indicated in this patient population (Oosterlinck et al., 2005). If hematuria is present, however, upper tract imaging is indicated for its evaluation regardless of the finding of bladder tumor.
16. **e. Are not indicated in most cases.** The yield of random bladder biopsy in patients with low-grade low-stage bladder tumors and a negative cytology is very low and is not indicated unless high-risk features are present.
17. **e. Higher than 45%. Untreated CIS is very high risk (>50%) for progressing to muscle-invasive disease. Even patients with a complete response to intravesical BCG will experience progression in 30% to 40% of cases on longitudinal follow-up** (Sylvester et al., 2005).
18. **c. Maintenance therapy.** Maintenance therapy is the only proven scenario demonstrating reduction of tumor progression. Dose reduction studies appear to support similar benefit for lower doses, which are usually better tolerated if side effects are identified with full dosing. Interferon is incrementally beneficial in certain circumstances. Options d and e are both without scientific basis but are commonly recommended.
19. **c. Solitary LG Ta >3 cm.** According the 2016 AUA/SUO guidelines, cases of NMIBC. can be classified as low-, intermediate-, or high-risk tumors (Table 136.2). LG Ta with recurrence within 1 year, solitary LG Ta tumors larger than 3 cm, multifocal LG Ta tumors, HG Ta tumors smaller than 3 cm, and LG T1 tumors are all considered to be of intermediate risk.
20. **e. a and d.** Appropriate surveillance for high-risk tumors includes cystoscopy every 3 to 4 months for 2 years, semiannually for 2 years, and annually for life with cytology on the same schedule. The clock should be restarted with each recurrence. Upper tract

imaging should be done annually for 2 years and considered on a less frequent basis after that.

21. **e. b and d.** The clinical scenario described represents a grade 3 complication given the persistent high-grade fever. This patient should be started on treatment with isoniazid and rifampin for 3 to 6 months (choice b) along with vitamin B6 or pyridoxine. Furthermore, consultation with an infectious disease specialist (choice d) should be considered in this setting, particularly if symptoms are severe. This patient should also be provided with symptomatic management with anticholinergics and/or nonsteroidal anti-inflammatory drugs (NSAIDs), etc. The dose of BCG should be reduced to a half or one-third dose or stopped completely if there is any evidence of solid organ involvement.
22. **b. Fibroblast growth factor receptor 3 (FGFR3).** Fibroblast growth factor receptor 3 is a commonly mutated cellular proliferation gene that is mutated in low-grade NMIBC. RB and p53, along with a number of mutations on chromosome 9, are commonly associated with high-grade tumors.
23. **c. 30% to 50%.** For patients who fail initial BCG, a second course provides a 30% to 50% chance of response.

Chapter Review

1. Malignant tumors are classified as low grade or high grade, regardless of invasion.
2. High- and low-grade cancers are often considered essentially separate diseases based on disparate genetic development, biologic behavior, and management strategies.
3. The most important risk factor for progression is grade.
4. CIS is a precursor as well as a risk factor for progression, invasion, and metastasis.
5. Papillary tumors with orderly cellular arrangement, minimal architectural abnormalities, and minimal nuclear atypia are designated papillary urothelial neoplasm of low malignant potential (PUNLMP).

6. TURBT is performed both to remove all visible tumors and provide specimens for pathologic examination to determine stage and grade.
7. Repeat resection within 2 to 6 weeks is usually indicated in patients with high-grade disease, especially if no muscle was present in the initial TURBT.
8. All suspicious lesions should be sampled, but random biopsies are not required in low-risk patients.
9. Office-based fulguration and observation may be applied to certain low-risk patients.
10. Fluorescence cystoscopy with 5-aminolevulinic acid (5-ALA) derivatives and narrow-band imaging improves the ability to visualize inconspicuous tumors and could reduce short- and long-term recurrence rates after TUR.
11. Single-dose intravesical chemotherapy administered within 6 hours of resection reduces recurrence of low-risk tumors, with significant impact in the setting of initial presentation of solitary low-grade papillary tumors.
12. The incremental benefit in patients with recurrent or multiple tumors is limited.
13. No benefit has been found in patients with high-grade disease.
14. Intravesical BCG has higher efficacy and side effects compared to intravesical chemotherapy.
15. BCG should be used cautiously for patients with low-risk disease because of concern about side effects.
16. BCG is the only agent shown to delay or reduce high-grade tumor progression.
17. The optimal dose and the treatment schedule for BCG are undetermined, but results are better with maintenance therapy, if tolerated.
18. BCG is contraindicated in the setting of a disrupted urothelium because of the risk of intravasation and BCG sepsis.
19. Intravesical chemotherapy has a clear impact on tumor recurrence when immediately instilled after TURBT and in the adjuvant setting. There is no clear evidence of an impact on progression.
20. Combinations of various chemotherapeutic agents and chemotherapy combined with BCG have not demonstrated major benefit combined

with single-agent treatment with the exception of interferon.

21. In general, side effects of chemotherapy tend to be less common and less severe than those for BCG, but BCG is more efficacious.
22. Combination intravesical chemotherapy may have a role in treating BCG-refractory cancers.
23. Patients who experience failure of an initial course of intravesical therapy after TURBT are at high risk for recurrence or progression.
24. Failure of initial chemotherapy or BCG is most appropriately treated with a subsequent course of BCG because its efficacy in this setting is significantly greater than that of chemotherapy.
25. Following failure of a second course of intravesical therapy, patients can be enrolled in clinical trials of new agents, consider combination intravesical therapy, or proceed to cystectomy.
26. Patients at high risk for progression should be considered for cystectomy.
27. Failure to respond to an initial course of intravesical therapy is occasion to reconsider cystectomy.
28. Failure to respond to a second course is an indication for immediate cystectomy unless contraindicated or the patient chooses to pursue clinical trials or newer proven intravesical options if evidence builds for their use.
29. Cystoscopy is the hallmark of surveillance. The surveillance schedule should be individualized on the basis of risk stratification of the most recently resected tumor.
30. A number of tumor markers have shown the ability to improve the sensitivity of cytology, but specificity is lower for most.
31. Increased fluids, smoking cessation, and a low-fat diet are recommended.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition*, on the Expert Consult website.

137

Management of Muscle-Invasive and Metastatic Bladder Cancer

Thomas J. Guzzo, John Christodouleas, and David J. Vaughn

Keywords

Bladder cancer; metastatic bladder cancer; muscle-invasive bladder cancer

Questions

1. Important components in the clinical staging of patients with muscle-invasive bladder cancer include which of the following?
    a. Transurethral resection with adequate detrusor muscle in the specimen
    b. Bimanual examination under anesthesia
    c. Cross-sectional imaging of the abdomen and pelvis
    d. Laboratory studies including liver function tests
    e. All of the above
2. The most important pathologic predictor of outcome following radical cystectomy for muscle-invasive bladder cancer is
    a. pT2a versus pT2b substaging.
    b. soft tissue margin status.
    c. nodal metastasis.
    d. ureteral margin status.
    e. prostatic urethral involvement.

3. All of the following have been reported to provide prognostic information with regard to pelvic lymphadenectomy EXCEPT
    a. absolute number of lymph nodes removed.
    b. laterality of a single positive lymph node.
    c. lymph node density.
    d. extranodal extension.
    e. anatomic extent of the lymph node dissection.
4. A 52-year-old male with cT2N0Mx urothelial carcinoma is undergoing a radical cystoprostatectomy and planned orthotopic urinary diversion. What intraoperative finding would be an absolute contraindication to orthotopic diversion?
    a. Positive ureteral frozen section for carcinoma in situ (CIS)
    b. Positive apical urethral frozen section for urothelial carcinoma
    c. Suspicious lymphadenopathy
    d. Greater than 1.5-L blood loss during the cystectomy
    e. None of the above
5. All of the following are true statements regarding neoadjuvant chemotherapy for muscle-invasive bladder cancer EXCEPT which?
    a. Meta-analysis of available randomized trial data has reported an absolute 5% survival advantage for patients who receive neoadjuvant chemotherapy.
    b. Neoadjuvant chemotherapy is likely underutilized in the US-based studies that use administrative data sets.
    c. Patients who are pT0 on final pathology following neoadjuvant chemotherapy have excellent oncologic outcomes.
    d. In patients in whom a cisplatin-based regimen is contraindicated, carboplatin-based chemotherapy provides similar oncologic efficacy.
    e. All of the above.
6. A 65-year-old woman with normal renal function undergoes up-front radical cystectomy, extended pelvic lymphadenectomy, and ileal conduit urinary diversion for bacille Calmette-Guérin–refractory bladder carcinoma in situ (CIS). Final pathology is notable for T2N1M0 disease. The next step in management should be
    a. enrollment in a clinical vaccine trial.

b. adjuvant cisplatin-based chemotherapy.
c. adjuvant pelvic external beam radiation therapy.
d. combination chemotherapy/external beam radiation therapy.
e. adjuvant carboplatin-based chemotherapy.

7. All of the following are contraindications to trimodal bladder preservation EXCEPT
   a. a solitary, completely resected tumor.
   b. hydronephrosis.
   c. diffuse bladder CIS.
   d. T3 disease on cross-sectional imaging.
   e. multifocal tumors.
8. Predictors of a poor response to chemotherapy in patients with locally advanced or metastatic bladder cancer include
   a. Karnofsky performance status below 80%.
   b. visceral metastasis.
   c. both a and b.
   d. neither a nor b.
9. Gemcitabine/cisplatin systemic therapy is often used in preference to MVAC (methotrexate, vinblastine, doxorubicin, and cisplatin) for patients with locally advanced and metastatic bladder cancer because of
   a. studies demonstrating improved progression-free survival.
   b. studies demonstrating improved overall survival.
   c. a better toxicity profile.
   d. oral administration compared with intravenous administration.
   e. cost.
10. An orthotopic neobladder in a woman undergoing anterior pelvic exenteration for muscle-invasive bladder cancer is contraindicated in the setting of
    a. age older than 75 years.
    b. nodal metastasis.
    c. recurrent urinary tract infection.
    d. a serum creatinine of 1.5.
    e. tumor invading the anterior vaginal wall.

Answers

1. **e.** Important components of staging patients with muscle-invasive bladder cancer include adequate transurethral resection, bimanual examination under anesthesia to assess local extent of disease, cross-sectional imaging, and serum laboratory values.
2. **c. Nodal metastasis. Approximately 25% of patients with clinical T2 disease will have lymph node metastasis at the time of radical cystectomy.** Lymph node status following surgery is a powerful predictor of long-term recurrence-free and overall survival. **Of patients with lymph node involvement, 70% to 80% will ultimately experience a recurrence following radical cystectomy.**
3. **b. Laterality of a single positive lymph node.** Although some surgical series have demonstrated improved survival in patients with one positive lymph node compared with those with multiple positive nodes, the laterality of a single positive lymph node has no prognostic significance.
4. **b. Positive apical urethral frozen section for urothelial carcinoma.** The only contraindication to performing an orthotopic neobladder is a positive apical urethral margin and inability to achieve a negative margin of the retained urethra.
5. **d. In patients in whom a cisplatin-based regimen is contraindicated, carboplatin-based chemotherapy provides similar oncologic efficacy. Although carboplatin is a reasonable choice for patients in whom cisplatin is contraindicated, it should not be considered a first-line therapy.** Patients who cannot undergo neoadjuvant cisplatin-based chemotherapy should be considered for immediate cystectomy.
6. **b. Adjuvant cisplatin-based chemotherapy.** Adjuvant cisplatin-based combination chemotherapy should be considered. Randomized trials have thus far not been definitive in overall survival results on this subject. A meta-analysis suggests a 9% absolute benefit in overall survival, but the trials often closed early

or because of poor accrual, a small number of patients, and inability to represent all of the patients in adjuvant chemotherapy trials.

7. **a. A solitary completely resected tumor. Patients with solitary completely resected tumors are ideal candidates for bladder preservation.**
8. **c. Both a and b. The Memorial Sloan Kettering Cancer Center group published their data on 203 patients with unresectable or metastatic bladder cancer treated with MVAC (). They found a Karnofsky performance status below 80% and visceral (lung, liver, bone) metastasis to be independent predictors of poor outcome.** Median survival times for patients who had zero, one, or two risk factors were 33, 13.4, and 9.3 months, respectively.
9. **c. A better toxicity profile.** The toxicity of MVAC led to trials of alternative, less toxic chemotherapy regimens. **Most notably, a phase III randomized trial comparing gemcitabine/cisplatin with MVAC was conducted in 405 patients** (von der Maase et al., 2000, 2005). [a] There was no difference in response rates (49% vs 46%), time to progression (7.4 vs 7.4 months), and overall survival rates (13.8 vs 14.8 months) between the two study arms. **The updated study analysis confirmed equivalence of the two regimens (hazard ratio, 1.09, 95% confidence interval, 0.88 to 1.34, *P* = .66). The gemcitabine/cisplatin regimen was better tolerated, with only 37% of patients in that arm requiring dose modifications compared with 63% in the MVAC arm.** Patients in the gemcitabine/cisplatin arm also experienced less grade 3/4 neutropenia, neutropenic fever, neutropenic sepsis, and mucositis. The toxicity-related death rate was also lower in the gemcitabine/cisplatin group (1% vs 3%). Because of its equivalent efficacy and better tolerability, gemcitabine/cisplatin is the most widely used chemotherapeutic regimen for muscle-invasive and metastatic bladder cancer.
10. **e. Tumor invading the anterior vaginal wall.** The distal two-thirds of the female urethra may serve as an adequate sphincter mechanism provided that the risk of cancer in the retained urethra is low. Anterior vaginal wall involvement—by a posterior-based

bladder tumor or bladder neck or urethral involvement—is a contraindication to urethral sparing and orthotopic bladder replacement because one cannot get an adequate distal vaginal margin and urethral margin (Stein et al., 1998). Age is not a contraindication as long as there is good pelvic support minimizing the risk of stress incontinence and the patient is capable of intermittent catheterization should the need arise. Nodal metastasis is associated with a 15% local recurrence rate with only a modest risk of invasion of the neobladder, and a thorough node dissection minimizes this risk (Lerner, 2009). Bilateral hydronephrosis, although indicating a deeply invasive cancer, is not a de facto contraindication (Stimson et al., 2010).

Chapter Review

1. Among those with muscularis propria–invasive bladder cancer, 80% are seen with the disease at initial presentation.
2. Deaths due to bladder cancer invariably occur as a result of distant metastases present at the time of local regional therapy and usually occur within the first 2 years following treatment. Therefore muscularis propria–invasive bladder cancer should be considered a systemic disease.
3. The micropapillary variant is an aggressive disease and does not respond particularly well to chemotherapy.
4. T1 grade 3 bladder tumors should routinely be re-resected because understaging is not an uncommon event.
5. Fat can be observed in the bladder wall and should not be confused with perivesical fat.
6. Lymphatic and vascular invasion are risk factors for metastases.
7. Following chemotherapy, metastases may occur in unusual locations, such as the central nervous system.
8. The incidence of pelvic node metastases is directly related to the depth of invasion and the presence of lymphovascular invasion.
9. As many as 50% of patients with muscularis propria–invasive bladder cancer succumb to their disease.

10. Of the randomized trials evaluating neoadjuvant therapy, most have not shown a definite survival advantage; however, a meta-analysis has shown a small survival advantage for those receiving neoadjuvant chemotherapy. The evidence for adjuvant chemotherapy conferring a survival advantage is less convincing.
11. Appropriate candidates for bladder preservation (transurethral tumor resection, chemotherapy, and radiation therapy thereby preserving the bladder) are those who have a solitary T2 lesion of small diameter with no associated hydronephrosis and a visibly complete resection. Such patients should have normal renal function.
12. As many as 15% of patients with tumors invading the muscularis propria have no residual disease following transurethral tumor resection.
13. Factors that affect outcome include stage, performance, status, lymphovascular invasion, age, gender, and histology.
14. Predictors of a poor prognosis include poor performance status and the presence of visceral metastases.
15. Treatment of neuroendocrine bladder tumors includes neoadjuvant chemotherapy and surgical resection; neuroendocrine tumors may be associated with a paraneoplastic syndrome (hypercortisolism and hypercalcemia).
16. There is a significant survival benefit to those who are rendered P0 at the time of radical cystectomy.
17. The highest concentration of prostatic ducts are located from the mid-prostate to the veru at the 5 and 7 o'clock positions and are the locations where CIS of the prostatic urethra is most likely to be found (biopsy should be performed in this location).
18. In women, bladder neck biopsies are a good surrogate for urethral biopsies when an orthotopic bladder is being considered.
19. Skip metastases in nodal disease for bladder cancer are rarely seen; this is not true of prostate cancer.
20. Ureteral margin status is a predictor of upper tract recurrence.
21. Prostatic stromal invasion carries a high risk of recurrent disease.
22. Patients who are not good candidates for cisplatin chemotherapy are those with a poor performance status, a creatinine clearance less than 60 mL/min, hearing loss, peripheral neuropathy, and heart failure.

23. Although carboplatin is a reasonable choice for patients in whom cisplatin is contraindicated, it should not be considered a first-line therapy.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

138

Surgical Management of Bladder Cancer

Transurethral, Open, and Robotic

Neema Navai, and Colin P.N. Dinney

Keywords

Bladder cancer; bladder tumor; laparoscopic surgery; minimally invasive; partial cystectomy; robotic surgery; surgery; transurethral resection

Questions

1. The administration of neoadjuvant chemotherapy has improved survival in muscle-invasive bladder cancer from:
    a. 16 to 42 months.
    b. 23 to 54 months.
    c. 37 to 51 months.
    d. 46 to 77 months.
    e. 75 to 85 months.
2. Upper-tract imaging for urothelial carcinoma may include all of the following EXCEPT:
    a. renal ultrasound.
    b. computed tomography (CT) abdomen and pelvis.
    c. whole-body positron emission tomography (PET)/CT.

d. magnetic resonance imaging (MRI) abdomen and pelvis.
e. retrograde pyelogram.

3. MRI-based contrast agents are absolutely contraindicated at which glomerular filtration rate (GFR) level?
   a. <15 mL/min
   b. <20 mL/min
   c. <30 mL/min
   d. <35 mL/min
   e. <60 mL/min
4. The improvement in 5-year survival and median survival when more than 10 lymph nodes are removed is approximately:
   a. 15%, 24 months.
   b. 10%, 36 months.
   c. 20%, 24 months.
   d. 5%, 15 months.
   e. 15%, 15 months.
5. Which of the following statements is TRUE?
   a. Urethral recurrence following radical cystectomy is approximately 8% at 5 years.
   b. Even patients with a negative intraoperative urethral frozen section are at high risk for recurrence.
   c. The negative predictive value of urethral frozen section is poor.
   d. Orthotopic neobladder is protective against urethral recurrence and therefore a positive urethral margin is not a contraindication.
   e. Orthotopic neobladder can only be performed after nerve-sparing radical cystectomy.
6. Which of the following statements is FALSE regarding nerve-sparing radical cystectomy?
   a. A technique analogous to radical prostatectomy is used.
   b. Sexual function is similar for capsular-sparing and conventional nerve-sparing techniques.
   c. Age is a strong predictor of the return of erectile function.
   d. Nerve sparing does not increase local recurrence rates.

e. Ejaculatory function can be maintained with subtotal prostate resection.

7. Anterior pelvic exenteration includes removal of the following EXCEPT:
    a. uterus.
    b. cervix.
    c. ovaries.
    d. urethra.
    e. vaginal introitus.
8. Partial cystectomy is appropriate in which of the following settings?
    a. 4-cm T2 lesion in the trigone
    b. 1-cm T2 lesion in the dome
    c. 3-cm T2 lesion in the dome with carcinoma in situ (CIS) in one location
    d. 1-cm T2 lesion with pelvic lymphadenopathy on imaging
    e. CIS in two locations
9. Enhanced recovery includes all of the following EXCEPT:
    a. alvimopan.
    b. neostigmine.
    c. pharmacologic thromboembolism prophylaxis.
    d. nasogastric suction.
    e. early enteral feeding.
10. Thromboembolism prophylaxis is needed:
    a. immediately before incision.
    b. postprocedure for 1 day.
    c. postprocedure for 1 week.
    d. postprocedure for 1 month.
    e. both a and d.
11. The RAZOR trial demonstrated which of the following:
    a. Inferior 2-year progression-free survival for open radical cystectomy
    b. Inferior 2-year progression-free survival for robotic radical cystectomy
    c. Superior oncologic outcomes for robotic radical cystectomy
    d. Noninferior 5-year progression-free survival for robotic radical cystectomy

e. None of the above

12. Which of the following statements is supported by level 1 evidence?
    a. Robotic radical cystectomy with minimally invasive urinary diversion has a similar complication rate to an open approach.
    b. Robotic radical cystectomy results in decreased blood loss when compared to open radical cystectomy.
    c. Robotic radical cystectomy and open urinary diversion is oncologically inferior to an open approach.
    d. Robotic radical cystectomy and minimally invasive urinary diversion is more costly than an open approach.
    e. None of the above.

Answers

1. **d. 46 to 77 months.** In a seminal randomized trial, Grossman and colleagues compared the treatment of muscle-invasive bladder cancer with radical cystectomy alone or surgery followed by three cycles of MVAC chemotherapy (methotrexate, vinblastine, doxorubicin, and cisplatin). They demonstrated a significant improvement in survival (46 vs. 77 months) in the neoadjuvant chemotherapy arm. This study serves as the basis for current treatment paradigms in muscle-invasive bladder cancer (Grossman et al., 2003). [a]
2. **c. Whole-body positron emission tomography (PET)/CT.** Conventional staging evaluation for upper-tract urothelial carcinoma should include evaluation of both the kidney parenchyma and the urothelial lumen. Although PET/CT can be useful for a staging evaluation, the resolution of imaging within the urinary tract is limited by the excretion of contrast material and lack of granular resolution.
3. **c. <30 mL/min.** Although gadolinium contrast should be administered with caution in patients whose GFR is between 30 and 60 mL/min, it is absolutely contraindicated in those with GFR <30 mL/min. This is due to the risk of nephrogenic systemic sclerosis.

4. **e. 15%, 15 months. In a study of surgical factors that influence outcomes in bladder cancer treatment, Herr and colleagues found that a lymph node dissection inclusive of more than 10 nodes was associated with improvement in survival of 15 months** (Herr et al., 2004).
5. **a. Urethral recurrence following radical cystectomy is approximately 8% at 5 years.** Factors that influence the risk of recurrence after radical cystectomy include orthotopic substitution with a positive urethral margin on frozen section analysis. This should be considered a contraindication for such a diversion. In addition, the negative predictive value is useful in the evaluation of urethral margins, and the risk of recurrence is only 8% at 5 years (Freeman et al., 1996).
6. **b. Sexual function is similar for capsular-sparing and conventional nerve-sparing techniques. The rate of natural potency after radical cystectomy with conventional nerve sparing is lower than that of analogous prostatectomy series. Studies examining sexual function after subtotal resection (e.g., prostate sparing) have demonstrated improved results (Spitz et al., 1999); however, caution is advised because of the high risk of concurrent occult prostate cancer and potential for increased local recurrence.**
7. **e. Vaginal introitus.** The vaginal introitus should be maintained for routine anterior exenteration. Satisfactory vaginal capacity can be maintained with both non–vaginal-sparing and vaginal-sparing approaches. In neither instance should a colpocleisis be performed as a matter of routine.
8. **b. 1-cm T2 lesion in the dome. In the setting of muscle-invasive bladder cancer, partial cystectomy can be considered in very select patients. In those with small lesions and a lack of concurrent CIS, the results of partial cystectomy approach those of radical cystectomy** (Kassouf et al., 2006; Capitanio et al., 2009).
9. **d. Nasogastric suction.** Postoperative nasogastric suction should be considered in patients with compromised airway protection; however, this has not been demonstrated to enhance recovery and

need not be incorporated to facilitate return of bowel function postoperatively. **Early enteral feeding, neostigmine, and alvimopan have all demonstrated efficacy in improving return of bowel function following abdominal surgery.**

10. **e. Both a and d.** In addition to the administration of prophylaxis prior to incision, a reduction in postoperative thromboembolic events from 4.6% to 0.8% was observed in patients treated for 4 weeks following abdominal or pelvic surgery (Kakkar et al., 2010).
11. **e. None of the above.** The RAZOR trial was designed as a prospective multiinstitutional noninferiority trial and demonstrated non-inferior progression-free survival at 2 years for robotic radical cystectomy when compared to open radical cystectomy (Parekh et al., 2018).
12. **b. Robotic radical cystectomy results in decreased blood loss when compared to open radical cystectomy.** The results of the RAZOR trial (Parekh et al., 2018) and the randomized study performed at Memorial Sloan Kettering Cancer Center (Bochner et al., 2014) demonstrated improvements in blood loss for robotic radical cystectomy. Notably both studies utilized an open technique for the urinary diversion.

Chapter Review

1. Before endoscopic treatment of bladder cancer, the patient should have upper-tract imaging.
2. Initial transurethral resection of a bladder tumor should routinely be performed to include muscle. There should be a 2-cm visibly negative margin on the surface.
3. Immediately following transurethral resection of bladder tumors, intravesical installation of epirubicin or mitomycin C modestly reduces recurrences but has little effect on progression.
4. Bacille Calmette-Guérin should never be instilled immediately following bladder tumor resection.
5. Before a cystectomy, the site of the abdominal stoma should be marked by an enterostomal therapist with the patient awake so that

the proper location may be ascertained.

6. If prostate- or prostate capsule–sparing techniques are to be used in orthotopic bladder construction, preoperative evaluation to rule out occult cancer—either transitional cell or prostate adenocarcinoma—should be performed.
7. A radical cystectomy in the female includes complete removal of the urethra including the meatus.
8. Patients amenable to partial cystectomy should have a solitary lesion without associated CIS in which a 2-cm margin may be obtained, which is far enough away from the ureteral orifices and bladder neck that closure can be accomplished without compromising these structures.
9. Sixty-four percent of patients undergoing radical cystectomy have at least one perioperative complication in the first 3 months postoperative; 13% experience high-grade complications. The majority of complications are gastrointestinal.
10. The boundaries of a standard lymph node dissection are the genitofemoral nerve laterally, internal iliac artery medially, Cooper ligament caudally, and crossing of the ureter at the common iliac artery cranially.
11. The 90-day mortality rate for radical cystectomy is approximately 3%.
12. Routine administration of antibiotic prophylaxis in patients undergoing a transurethral resection of a bladder tumor is recommended and should be given 30–60 minutes before the procedure.
13. Transurethral resection of tumors on the lateral wall may initiate the obturator reflex and result in bladder perforation. This may be minimized by minimally distending the bladder, using bipolar cautery, and using general anesthesia with muscle paralysis.
14. Routine stenting of a resected ureteral orifice with cutting current is not necessary.
15. In preparation for a cystectomy and urinary diversion, mechanical and antibiotic bowel prep is controversial. The data to justify omitting this preparation come, for the most part, from the general surgical literature in which the bowel is not opened in the peritoneal

cavity as it is in urology, particularly in continent diversions. Administration of intravenous antibiotic prophylaxis 30–60 minutes before the incision is recommended.

16. In women, the vagina should be prepped into the surgical field.
17. Care should be taken when using sealing instruments near the rectum because that organ may be injured by radiating heat.
18. Early enteral feeding, neostigmine, and alvimopan have all demonstrated efficacy in improving return of bowel function following abdominal surgery.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

139

Use of Intestinal Segments in Urinary Diversion

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Keywords

bowel anastomosis; enhanced recovery; ileal conduit; orthotopic neobladder; ureteral anastomosis; urinary diversion

Questions

1. When a portion of stomach is to be used for augmentation, it should:
    a. always be based on the right gastroepiploic artery.
    b. include only the antrum.
    c. never extend to the pylorus.
    d. include a significant portion of the lesser curve.
    e. be mobilized with the omentum.
2. The ileum differs from the jejunum in that:
    a. it has a larger diameter.
    b. the mesentery is thinner.
    c. it has multiple arcades.
    d. the vessels in the mesentery are larger.
    e. the mesentery is longer.
3. When stomach is used for urinary diversion, the electrolyte abnormality that may occur is most commonly what type of metabolic alkalosis?

a. Hyperchloremic
b. Hypochloremic
c. Hyperkalemic
d. Hypernatremic
e. Hypocalcemic

4. Postoperative bowel obstruction is most common when which of the following segments is used for diversion?
   a. Right colon
   b. Stomach
   c. Sigmoid
   d. Ileum
   e. Transverse colon
5. Mechanical bowel preparation results in a reduction in:
   a. bacterial counts per gram of enteric contents.
   b. bacterial count in the jejunum.
   c. total number of bacteria in the bowel.
   d. bacterial counts in the stomach.
   e. bacterial counts in the ileum.
6. Systemic antibiotics in elective surgery should be given:
   a. before the patient is anesthetized.
   b. before the skin incision is made.
   c. intraoperatively before closure commences.
   d. at any time in the perioperative period.
   e. postoperatively for 3 to 5 days.
7. The most common cause of a lethal bowel complication is:
   a. use of previously irradiated bowel.
   b. lack of mechanical bowel prep.
   c. lack of antibiotic bowel prep.
   d. placement of a drain adjacent to the anastomosis.
   e. failure to give preoperative antibiotics.
8. When stapled anastomoses are compared with sutured anastomoses, there is/are:
   a. fewer leaks.
   b. less compatibility with urine.
   c. reduced overall operative time.
   d. lesser incidence of bowel obstruction.

e. earlier return of bowel function.

9. The use of a nasogastric tube in the postoperative period:
    a. hastens the return of intestinal motility.
    b. reduces the incidence of bowel leak.
    c. reduces postoperative vomiting.
    d. increases the risk of aspiration.
    e. reduces the incidence of anastomotic leak.
10. The abdominal stoma for a conduit should be:
    a. flush with the skin.
    b. placed through the belly of the rectus muscle.
    c. made as a loop to reduce parastomal hernia.
    d. made with the colon for the lowest complication rate.
    e. placed in the right lower quadrant.
11. The loop end ileostomy is best used in:
    a. the obese patient.
    b. the thin patient.
    c. when a stoma is revised.
    d. in female patients.
    e. in spinal cord injury patients.
12. Ureteral strictures occurring after an ileal conduit not associated with the ureteral intestinal anastomosis most frequently occur:
    a. at the ureteropelvic junction.
    b. in the right ureter several centimeters proximal to the ureteral intestinal anastomosis.
    c. on the left side where the ureter crosses the aorta.
    d. in the mid-ureter.
    e. in either ureter within several centimeters proximal to the anastomosis.
13. Renal deterioration after a conduit diversion with normal kidneys occurs in what percent of renal units?
    a. 20%
    b. 40%
    c. 50%
    d. 70%
    e. 80%

14. The most common cause of death in patients with ureterosigmoidostomies during the long term is:
    a. cancer.
    b. renal failure.
    c. acid–base abnormalities.
    d. the primary disease.
    e. ammonium intoxication.
15. The minimal glomerular filtration rate (GFR) in mL/min necessary for a continent diversion is:
    a. 70.
    b. 60.
    c. 35.
    d. 25.
    e. 20.
16. The urinary diversion with the fewest intraoperative and immediate postoperative complications is:
    a. ileal conduit.
    b. colon conduit.
    c. Koch pouch.
    d. Indiana pouch.
    e. neobladder.
17. The jejunal conduit syndrome is manifested by:
    a. hyperchloremic metabolic acidosis.
    b. hypochloremic metabolic alkalosis.
    c. hyperkalemic, hyponatremic metabolic acidosis.
    d. hypokalemic, hyponatremic metabolic alkalosis.
    e. hyperkalemic metabolic alkalosis.
18. The primary advantage of a transverse colon conduit is:
    a. its ease of construction.
    b. the ability to perform a non-refluxing anastomosis.
    c. less likely to be injured by radiation.
    d. reduced electrolyte problems.
    e. equidistant from each kidney, allowing for short ureteral length on both sides.
19. Total body potassium depletion is most common in:
    a. ureterosigmoidostomy.

b. ileal conduit.
c. colon conduit.
d. sigmoid conduit.
e. gastrocystoplasty.

20. In urinary intestinal diversion, serum creatinine may not be an accurate reflection of renal function because of:
    a. interfering substances.
    b. tubule secretion.
    c. tubule reabsorption.
    d. bowel reabsorption.
    e. decreased renal elimination.
21. Patients with urinary diversions who have a hyperchloremic metabolic acidosis with time:
    a. retain the ability to maintain the acidosis.
    b. lose the ability for electrolyte transport in the intestinal segments.
    c. compensate for the metabolic acidosis, thus eliminating risk.
    d. intermittently absorb ammonia when infection is present.
    e. tend to retain potassium.
22. Bone density abnormalities:
    a. are unlikely to occur with ileum.
    b. are most likely to occur with colon.
    c. are more common in patients with persistent hyperchloremic metabolic acidosis.
    d. are common in patients with total body potassium depletion.
    e. are unlikely to occur in patients with conduits.
23. Urinary intestinal diversion in children:
    a. increases the need for vitamin D.
    b. increases the need for calcium.
    c. limits linear growth.
    d. decreases epiphyseal growth.
    e. results in premature epiphyseal closure.
24. Cancer occurring in urinary intestinal diversion is most likely to occur in:
    a. augmentations.
    b. colon conduits.

c. ileal conduits.
d. ureterosigmoidostomies.
e. sigmoid conduits.

25. Reconfiguring the bowel during the long term results in:
    a. decreased motor activity.
    b. increased volume.
    c. decreased metabolic complications.
    d. decreased absorption of solutes.
    e. increased absorption of solutes.
26. The syndrome of severe metabolic alkalosis in patients who have had a gastrocystoplasty is most likely to occur in patients who have:
    a. decreased aldosterone levels.
    b. jejunum interposed in the urinary tract.
    c. total body potassium depletion.
    d. elevated gastrin levels.
    e. decreased renin levels.
27. There is direct evidence from cystectomy patients that the following can be safely omitted from an enhanced recovery after surgery (ERAS) protocol:
    a. preoperative fasting
    b. thrombosis prophylaxis
    c. mechanical bowel prep
    d. pelvic drain
    e. preoperative intravenous antibiotics
28. Arginine hydrochloride infusion can be used to treat life-threatening complications caused by the following type of conduit:
    a. stomach
    b. jejunum
    c. ileum
    d. colon
    e. all of the above

Answers

1. **c. Never extend to the pylorus.** When a wedge of fundus is used, it should not include a significant portion of the antrum and should never extend to the pylorus or all the way to the lesser curve of the stomach.
2. **c. It has multiple arcades. The ileum, more distal in location, has a smaller diameter. It has multiple arterial arcades,** and the vessels in the arcades are smaller than those in the jejunum.
3. **b. Hypochloremic.** Complications specific to the use of stomach include the hematuria-dysuria syndrome and uncontrollable metabolic alkalosis in some patients. **When stomach is used, a hypochloremic, hypokalemic metabolic alkalosis may ensue.**
4. **d. Ileum.** The incidence of postoperative bowel obstruction is 4% to 10%. Colon, stomach, and sigmoid obstruction result in a 4% incidence, less than that occurring with ileum.
5. **c. Total number of bacteria in the bowel.** The mechanical preparation reduces the amount of feces, whereas the antibiotic preparation reduces the bacterial count. **A mechanical bowel preparation reduces the total number of bacteria but not their concentration.**
6. **a. Before the patient is anesthetized. Systemic antibiotics must be given before the operative event if they are to be effective.**
7. **a. Use of previously irradiated bowel.** In one study of urinary intestinal diversion, 75% of the lethal complications that occurred in the postoperative period were related to the bowel. Eighty percent of these patients had received radiation before the intestinal surgery.
8. **b. Less compatibility with urine.** In general, anastomoses using reabsorbable sutures or reabsorbable staples are preferable for intestinal segments that are exposed to urine.
9. **c. Reduces postoperative vomiting.** In several studies there was no significant difference in major intestinal complications between those who had postoperative nasogastric tubes and those who did not; however, those who did not have gastric decompression showed a much greater incidence of abdominal distention, nausea, and vomiting.

10. **b. Placed through the belly of the rectus muscle. All stomas should be placed through the belly of the rectus muscle** and be located at the peak of the infraumbilical fat roll.
11. **a. The obese patient.** The loop end ileostomy is usually easier to perform than the ileal end stoma in the patient who is obese.
12. **c. On the left side where the ureter crosses the aorta.** Of importance is that ureteral strictures also occur away from the ureterointestinal anastomosis. This stricture is most common in the left ureter and is usually found as the ureter crosses over the aorta beneath the inferior mesenteric artery.
13. **a. 20%.** Patients who are studied during the long term show a significant degree of renal deterioration. Indeed, 20% of renal units have shown significant anatomic deterioration.
14. **b. Renal failure.** The most common cause of death in patients who have had a ureterosigmoidostomy for more than 15 years is acquired renal disease (i.e., sepsis or renal failure).
15. **b. 60. If the patient is able to achieve a urine pH of 5.8 or less, can establish a urine osmolality of 600 mOsm/kg or greater in response to water deprivation, has a GFR that exceeds 60 mL/min, and has minimal protein in the urine, he or she may be considered for a retentive diversion.**
16. **a. Ileal conduit.** It is the simplest type of conduit diversion to perform and is associated with the fewest intraoperative and immediate postoperative complications.
17. **c. Hyperkalemic, hyponatremic metabolic acidosis.** The early and long-term complications are similar to those listed for ileal conduit except that the electrolyte abnormality that occurs is hyperkalemic, hyponatremic metabolic acidosis instead of the hyperchloremic metabolic acidosis of ileal diversion.
18. **c. Less likely to be injured by radiation.** The transverse colon is used when one wants to be sure that the segment of conduit used has not been irradiated in individuals who have received extensive pelvic irradiation.
19. **a. Ureterosigmoidostomy.** Hypokalemia and total body depletion of potassium may occur in patients with urinary intestinal diversion. This is more common in patients with

ureterosigmoidostomies than it is in patients who have other types of urinary intestinal diversion.

20. **d. Bowel reabsorption. Because urea and creatinine are reabsorbed by both the ileum and the colon, serum concentrations of urea and creatinine do not necessarily accurately reflect renal function.**
21. **a. Retain the ability to maintain the acidosis.** The ability to establish a hyperchloremic metabolic acidosis appears to be retained by most segments of ileum and colon over time.
22. **c. Are more common in patients with persistent hyperchloremic metabolic acidosis. Osteomalacia in urinary intestinal diversion may be due to persistent acidosis, vitamin D resistance, and excessive calcium loss by the kidney.** It appears that the degree to which each of these contributes to the syndrome may vary from patient to patient.
23. **c. Limits linear growth.** There is considerable evidence to suggest that **urinary intestinal diversion has a detrimental effect on growth and development.**
24. **d. Ureterosigmoidostomies.** The highest incidence of cancer occurs when the transitional epithelium is juxtaposed to the colonic epithelium and both are bathed by feces.
25. **b. Increased volume.** Reconfiguring bowel usually increases the volume, but its effect on motor activity and wall tension over the long term is unclear at this time.
26. **d. Elevated gastrin levels. The syndrome of severe metabolic alkalosis is most likely to occur in patients with high resting gastrin levels who are dehydrated and fail to empty their pouch in a timely manner.**
27. **c. Mechanical bowel prep.** There is direct evidence for omission of a mechanical bowel can be safely omitted from cystectomy and urinary diversion patients on an ERAS pathway. Thrombosis prophylaxis (b.) and perioperative antibiotics (e.) should never be omitted. There is is evidence that preoperative fasting (a.) and a pelvic drain (d.) may be safely omitted in colorectal surgery patients, but this has not be studied in depth in cystectomy population.

28. **a. Stomach.** Only urinary diversion consisting of stomach result in metabolic alkalosis. Arginine hydrochloride infusion plays no role in the management of acidosis caused by other bowel segments.

Chapter Review

1. Perioperative care. The use of a preoperative mechanical bowel prep, oral antibiotic bowel prep, and postoperative nasogastric tube decompression in patients undergoing bowel surgery is controversial, and can be safely omitted in appropriately selected patients. Administering intravenous antibiotics 1 h before the surgical incision is not controversial and is supported by many studies. Patients undergoing elective intestinal surgery in the studies that show no advantage to a mechanical and/or antibiotic bowel prep received preoperative intravenous antibiotics. It should be appreciated that the majority of these studies involve isolated anastomoses—not large segments of bowel that are opened, as is the case in urologic procedures.
2. Ureteral intestinal anastomotic strictures. Antirefluxing anastomoses have a 10%–20% stricture rate; refluxing anastomoses have a 3%–10% stricture rate. The Wallace ureteral intestinal anastomosis has the lowest stricture rate.
3. Renal function and urinary diversion. Serum creatinine and blood urea nitrogen do not accurately reflect renal function in patients with intestine in the urinary tract because these substances, when excreted by the kidney, are reabsorbed by the bowel. This is more likely to be a problem in continent diversions. A glomerular filtration rate (GFR) of at least 60 mL/min and an ability to acidify the urine are necessary prerequisites for a continent diversion.
4. The electrolyte abnormality that occurs when ileum or colon are used for the diversion is a hyperchloremic metabolic acidosis (Table 139.1). These patients may have a potassium deficiency as well.
5. Significant perioperative infectious complications occur in up to 10% of patients undergoing cystectomy and urinary diversion.

6. The most common cause of mortality in urologic procedures when the gut is used relates to complications involving the bowel.
7. Complications specific to the use of stomach include the hematuria-dysuria syndrome and uncontrollable metabolic alkalosis in some patients.
8. The incidence of postoperative bowel obstruction is 4%–10%. Colon, stomach, and sigmoid obstruction result in a 4% incidence, less than that occurring with ileum.
9. The mechanical preparation reduces the amount of feces and therefore total bacteria, whereas the antibiotic preparation reduces the bacterial count and therefore the concentration of bacteria.
10. If the patient is able to achieve a urine pH of 5.8 or less, can establish a urine osmolality of 600 mOsm/kg or greater in response to water deprivation, has a GFR that exceeds 60 mL/min, and has minimal protein in the urine, he or she may be considered for a retentive diversion.
11. Osteomalacia in urinary intestinal diversion may be due to persistent acidosis, vitamin D resistance, and excessive calcium loss by the kidney.
12. In patients with gastric tissue in the urinary tract (usually gastrocystoplasty), the syndrome of severe metabolic alkalosis is most likely to occur in those with high resting gastrin levels who are dehydrated and fail to empty their pouch in a timely manner.

TABLE 139.1

Segment	Syndrome	BLOOD						
		Na^+	K^+	Cl^-	pH	Associated Abnormalities	Symptoms	Treatment
Stomach	Severe metabolic alkalosis	--	↓	↓	↑	Elevated aldosterone	Lethargy, muscle weakness, respiratory insufficiency, seizures, ventricular arrhythmia	H_2 blocker, proton pump inhibitor; if life-threatening, arginine hydrochloride infusion and/or removal of segment
Jejunum	Hyperkalemic, hypochloremic metabolic acidosis	↓	↑	↓	↓	Elevated renin and angiotensin	Lethargy, nausea, vomiting, dehydration, muscle weakness	IV hydration, sodium bicarbonate, thiazide; if life threatening, removal of segment
Ileum/Colon	Hyperchloremic metabolic acidosis	--	↓	↑	↓	Total-body potassium depletion, hypocalcemia	Fatigue, anorexia, lethargy, weakness	Potassium citrate, sodium citrate, citric acid, sodium bicarbonate, chlorpromazine, nicotinic acid